

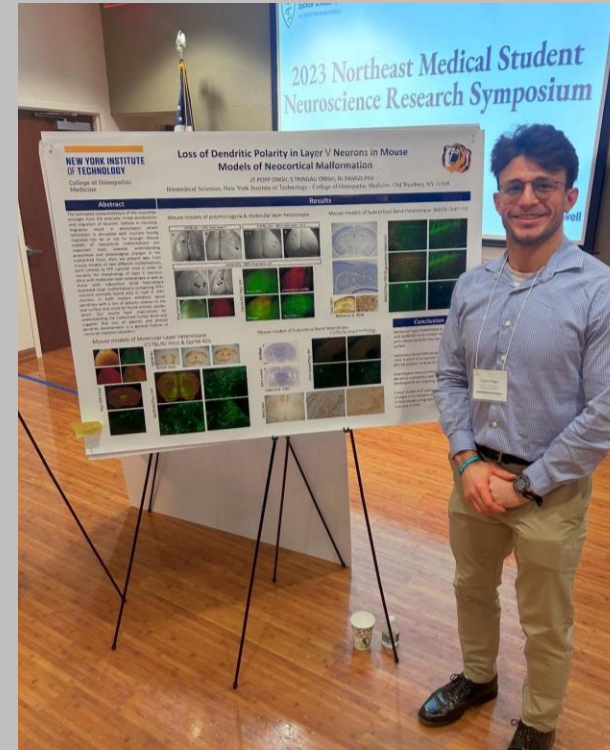
PPOM1 Class of 2029 - Seizures & Epilepsy

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

- 1) Students will apply the pathophysiology of seizures to distinguish between different types of seizure presentations.
- 2) Discover how genetic mutations can cause epilepsy syndromes.
- 3) The students will be informed about different types of epilepsy therapy approaches.

Seizures Throughout History (Not Tested)

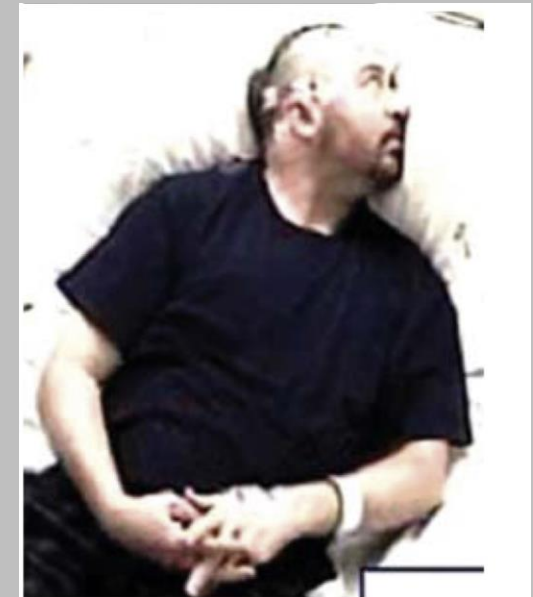
- The word "seizure" originates from the Latin word *sacire*, meaning "to take possession of" or "take by force". The word "seizure" was first used in the Middle English period (1150—1500).
- The word "epilepsy" comes from the Greek word *epilēpsía*, which comes from the verb *lambánein*, meaning "to seize". The word *epilambanein* means "to seize, possess, or afflict".
- Seizures and epilepsy in humans and the concept in general has been documented as early as 2500 BC. In the Homeric Era, people believed that epilepsy was a sacred disease induced by gods. In his work "On the Sacred Disease", Hippocrates challenged this perception of a godly origin of seizures. He proposed that seizures and epilepsy had natural or biologic causes.
- The pt undergoing trepanation (drilling into skull) and cauterization at simultaneously.
- Seizures have been apart of human culture and experience for thousands of years. A lot philosophy behind medical practice was superstition. Trepanation → to drain evil spirits
- People have been making noble yet rudimentary steps at trying to treat seizures and epilepsy for hundreds of years. Painting depicts a surgical approach.
- Careers in neurology with epileptology

- *Epilepticus Sic Curabitur* ("the way to cure an epilepticus") From Sloane Manuscript, collection of medical manuscripts, end of the 12th century. Author unknown. One of the first artistic descriptions of epilepsy surgery



Seizures Having Various Clinical Presentations

- Seizure is a clinical sign. Used for hundreds of years to help clinicians diagnose and try to treat it
- Different brain regions involved relating to the patient presentation
- Major Point: All things we experience; every variation of emotion, urge, desire or movement, sensation and perception has underpinning neuroanatomical location(s) that is/are intimately involved in the interpretation and manifestation of that specific experience.



Seizure: – clinical sign

- Transient alteration in CNS sensory, motor, autonomic, and/or cognitive function due to excessive & synchronous abnormal neuronal activity from a population of cells in the brain (neocortex, thalamus, etc).
- This excessive & synchronous abnormal neuronal activity leads to disruption in the homeostatic balance between excitation and inhibition.
- **Can have seizures but not have epilepsy, but cannot have epilepsy without seizures→ strict definition later slide**

Drugs, chemicals, compounds, etc reported to induce seizures

Alcohols and glycols	Cardiovascular agents	Mushrooms
Anesthetics, general	Drugs of abuse	Neuroleptics
*Antibiotics	Drug withdrawal	Neuromuscular blockers
Anticonvulsants	Hypoglycemics	NSAIDs
Anticholinergics	Hydrocarbons	Opioids
Antidepressants	Immunosuppressives	*Plants
Antifungals	Inhalants	*Radiographic contrast media
Antihistamines	Insecticides	Rodenticides
Antineoplastics	Metal Chelators	Sympathomimetics
Antiparasitics	Insect repellent	Vaccines (due to fever)
Antivirals	Metals	
Asphyxiants	Muscle relaxants	

• Don't memorize this



Causes of seizures are diverse

- head injury, acute metabolic or electrolyte changes, medications, stroke, tumor, fever, hydrocephaly, infection (neurocysticercosis, COVID-19), etc...)
- Can be **focal** (discretely localized to one hemisphere) or **generalized** (diffusely distributed across both hemispheres.)

VITAMINS – mnemonic for seizure cause

- V**ascular event
- I**nfection
- T**rauma
- A**utoimmune
- M**etabolic
- I**ngestion or withdrawal (ex. Alcohol)
- N**eoplasm
- pS**ych → EEG negative
- P**regnancy → Eclampsia

Common Types of Seizures: *behavioral profile*

- 1] **Tonic seizures**: stiffening of muscles of the body, generally those in the back, legs, and arms.
- 2] **Clonic seizures**: repeated jerking movements of muscles on one or both sides of the body.
- 3] **Myoclonic seizures**: twitches of limited body parts, arms, or legs (<https://www.youtube.com/watch?v=dU8R0LCCn6k>).
- 4] **Atonic seizures/"drop attacks"**: loss of muscle tone. The affected person will fall down or may drop his or her head involuntarily. Prone to head/musculoskeletal injury and need helmet.
- 5] **Tonic-clonic (Grand mal) seizures**: a mixture of symptoms, including stiffening of the body and repeated jerks of the arms and/or legs as well as loss of consciousness. Often preceded by sudden vocalization. Loss of bowel/bladder control during seizure. Sleepiness & confusion follow the seizure (<https://www.youtube.com/watch?v=Nds2U4CzvC4>).
- 6] **Absence seizures (Petit mal)**: the person may appear to be staring into space (change in consciousness). Often go right back to activity after the seizure. (<https://www.youtube.com/watch?v=H3iLQi6wt94>)
- 7] **Gelastic seizures** bouts of laughing/giggling without other convulsive seizures (<https://www.youtube.com/watch?v=x98yqTt754k>)
- 8] **Dacrystic seizures** bouts of crying with lacrimation. Often follow bouts of gelastic seizure.
- 9] **Emetic seizures**: convulsive seizure followed by a bout of nausea and vomiting.
- 7] **Psychogenic seizure**: episodes of movements(convulsions), sensations, or behaviors that are similar to epileptic seizures but do not have a neurologic origin (no signature EEG change)

More on Terminology/Experiential Phenomena

Convulsion

- Old and vague term typically used to denote a tonic-clonic seizure
- May be used to indicate a seizure with prominent motor activity

Automatism

- Repetitive motor activities that are coordinated and resemble a voluntary movement but are not purposeful (lip-smacking, chewing, swallowing, opening/closing fingers).
- This is not tardive dyskinesia!!! D2 receptor antagonism on cholinergic neurons in the striatum of the basal ganglia leads to aberrant and excessive acetylcholine release causing the features of TD. Pt who are manic, or psychotic will get antipsychotic medications in the ED setting, see on rotations.
- Many classifications of automatisms (**gelastic = bouts of laughing/giggling without other convulsive seizures; dacrystic = bouts of crying with lacrimation. Often follow bouts of gelastic seizure.**)

Sensory/subjective phenomena

- Visual (occipital lobe) and auditory (temporal lobe) hallucinations (Flickering/flashing lights, visual loss, spots in vision or hearing changes)
- Somatosensory (paresthesia's, shock-like sensations, numbness, pain, or a sense of movement or a desire to move a body part); Gustatory (e.g. metallic taste)

Experiential phenomena

- Fear, sadness, elation;
- feelings of familiarity (déjà vu) or unfamiliarity (jamais vu);

complex hallucination (such as seeing people or hearing music) and illusions (alterations of perception)

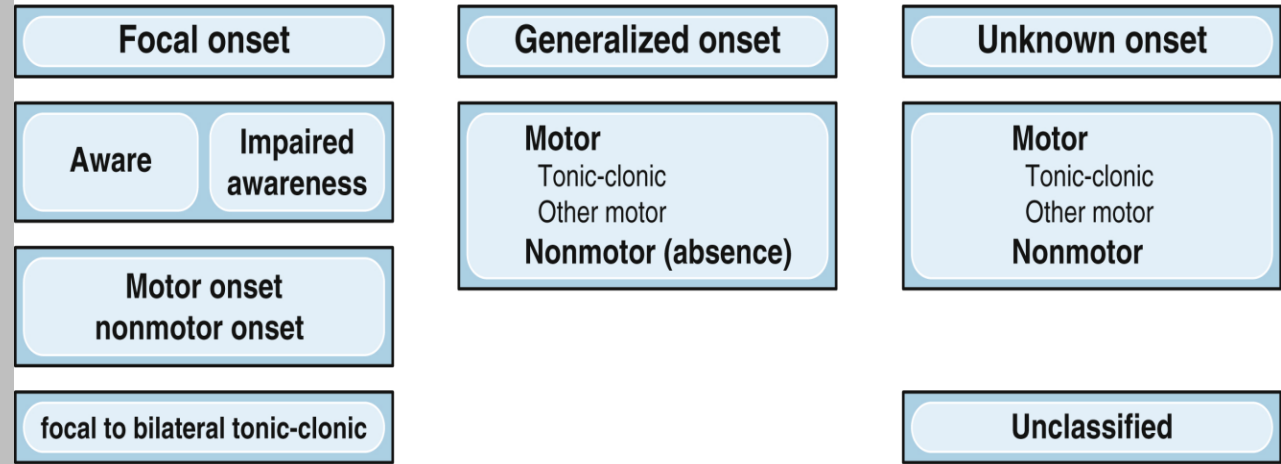
Classifying seizures...*always evolving with advances in medicine...*

Operational classification of seizure types by the International League Against Epilepsy: Position Paper of the ILAE Commission for Classification and Terminology. Fisher et al. 2017 Epilepsia

1] **Generalized seizures** are a result of abnormal neuronal activity **on both sides** of the brain & typically cause **loss of consciousness**, falls, or massive muscle spasms.

2] **Focal seizures** occur in **one hemisphere or one lobe in one hemisphere**. For example, someone might be diagnosed with focal frontal lobe seizures (based on EEG). Partial is old term as well as simple

ILAE 2017 classification of seizure types: basic version

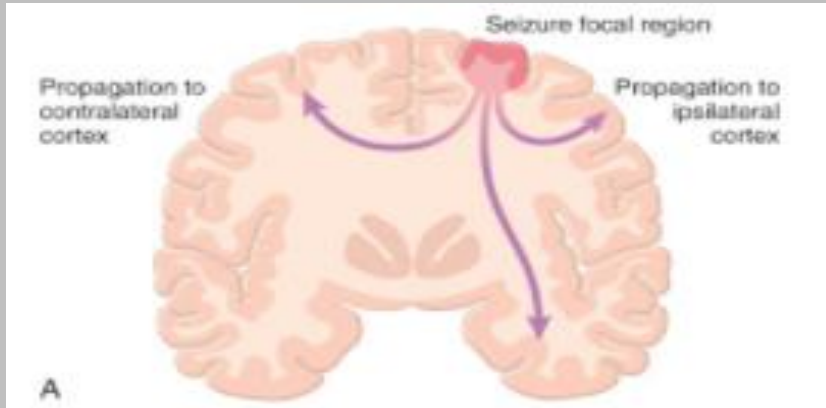


Old terminology

New terminology (know for exam)

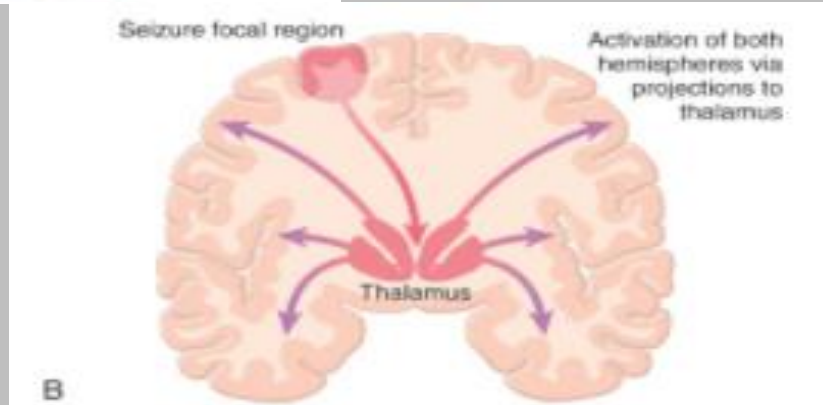
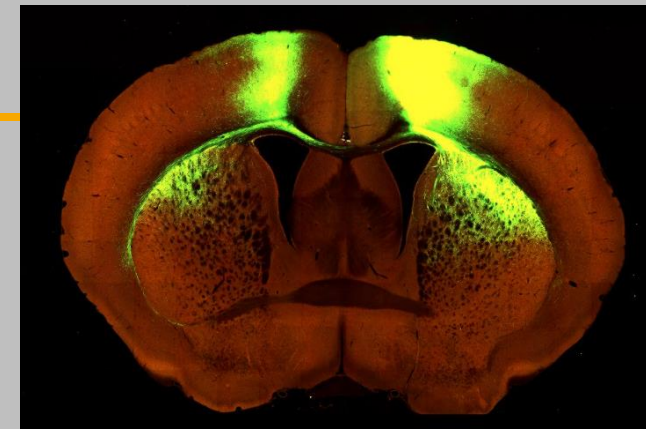
- Simple **Partial** → **Focal** aware
- Complex **Partial** → **Focal** Impaired awareness

Seizure Neuroanatomy:



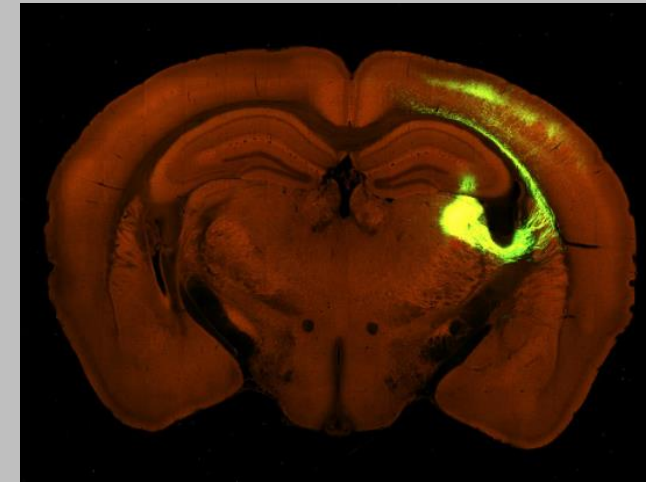
- Spread of epileptic activity through circuits in the brain

Cortic-cortical
Cortico-subcortical

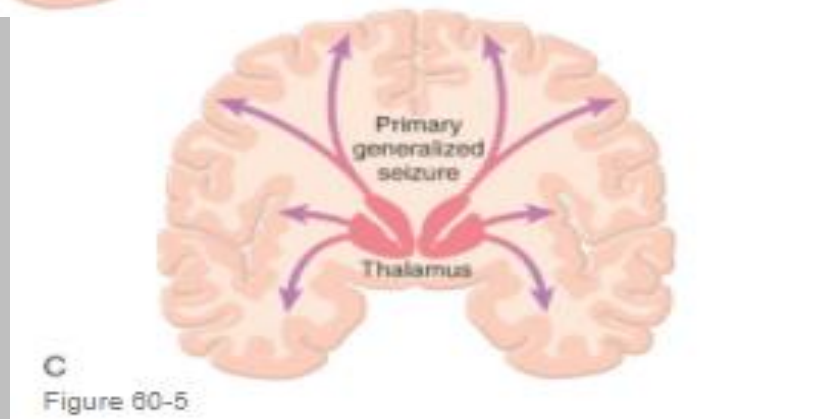


Thalamo-cortical

- Vital part of absence seizure pathophysiology



Guyton and Hall, Textbook of Medical Physiology



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Focal and Generalized Seizures in next slides..

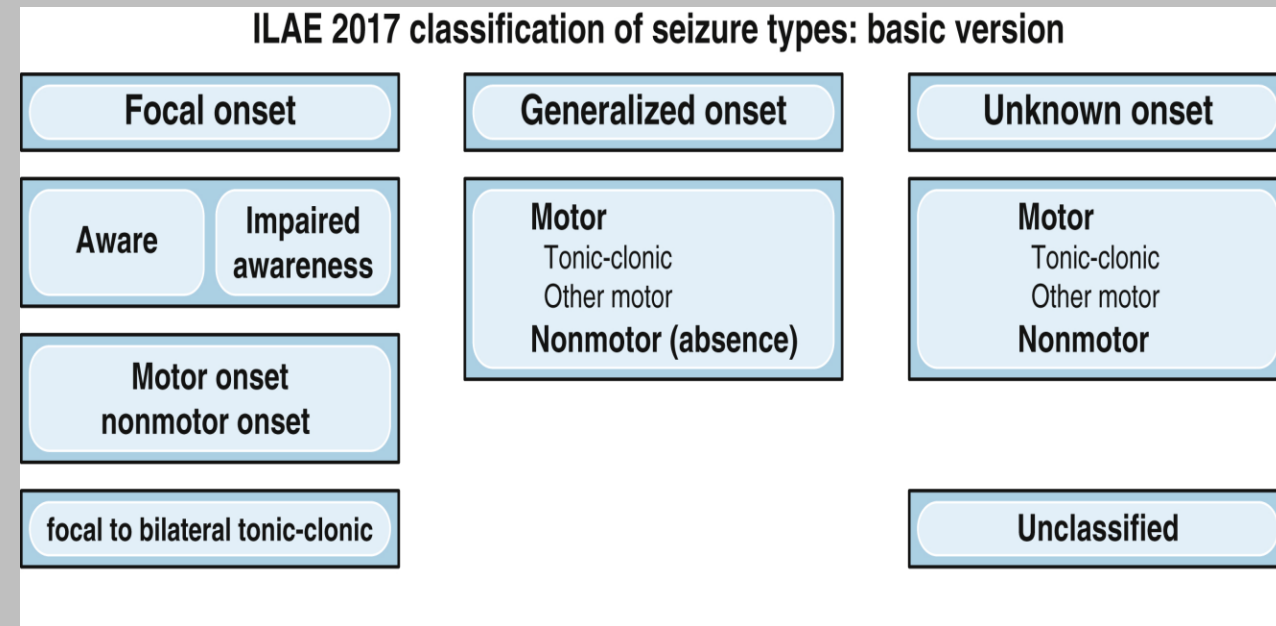


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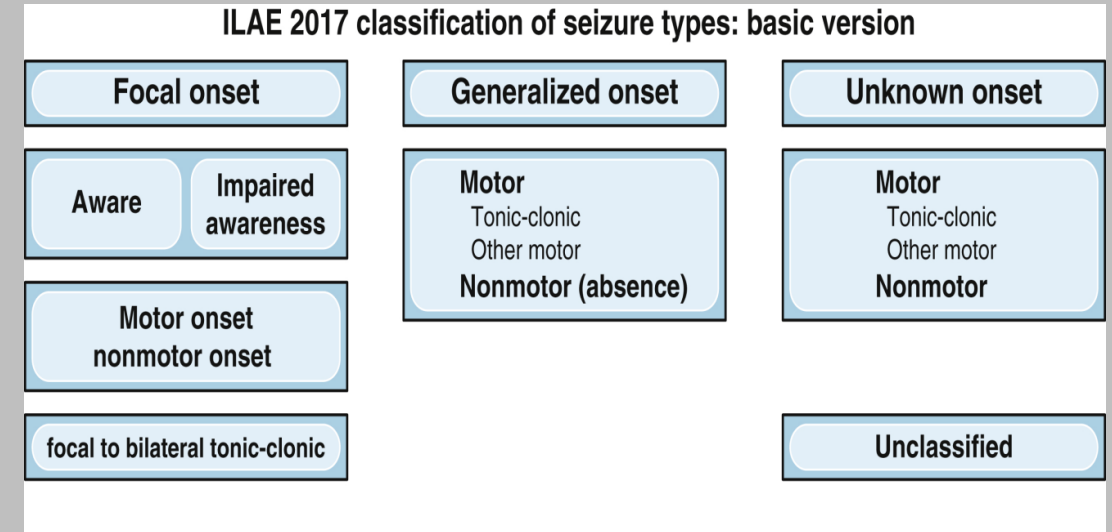
Focal Onset: Focal Aware Seizures

- Formerly called *Simple Partial seizures*
- *Focal* → *Originates on one hemisphere of the brain, can become generalized if seizure activity crosses corpus callosum or anterior commissure*
- *Automatisms can occur such as involuntary lip-smacking or blinking*
- **Awareness or mental status is not impaired**
- Focal aware seizures can have auras which are = subjective and experiential phenomena such as alterations of sensory perception (like taste and smell.) The manifestations depend on brain region involved
 - E.g. clonic movement of the left upper extremity indicates right pre-central gyrus seizure activity. Think of the homunculus!!

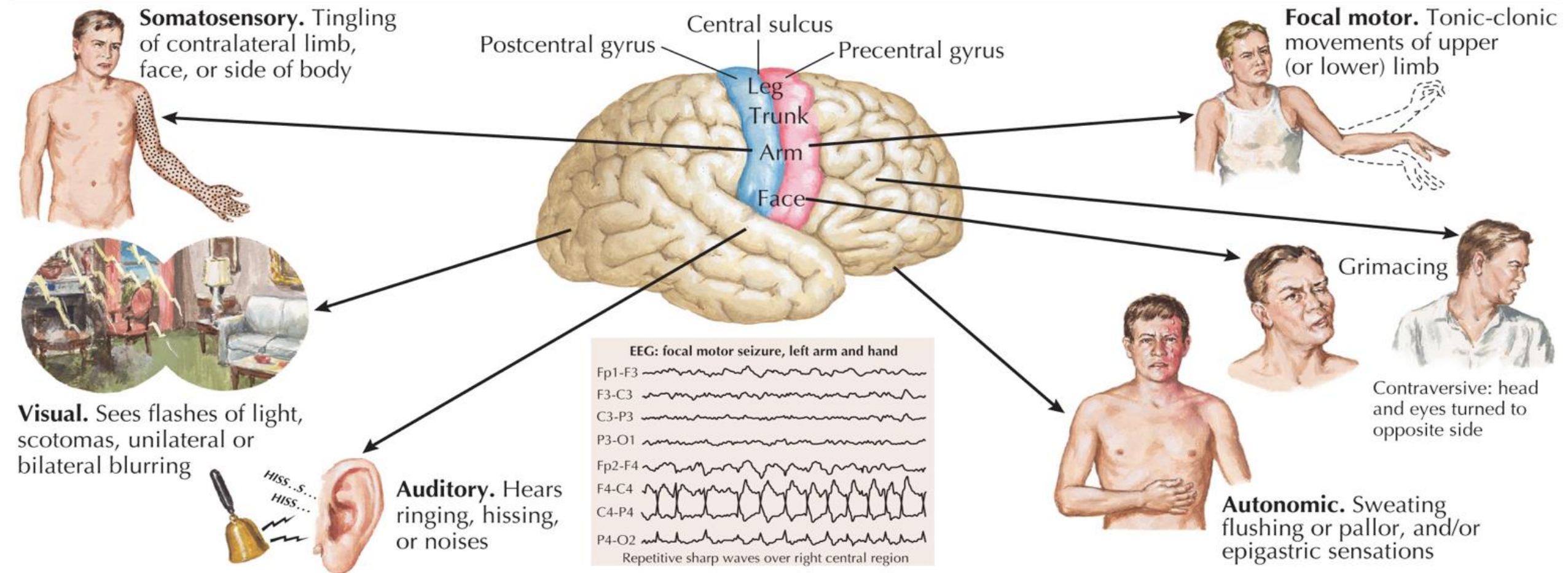


Focal Onset: Focal Impaired Awareness Seizure

- “FAIS”
- Formerly known as Complex Partial Seizures (complex = altered mentation)
- Characterized by altered awareness or mental status during the seizure
- May be very subtle, manifesting with slight confusion, or slowing of responses
- May experience amnesia after the event
- Sometimes difficult to determine if awareness was impaired.
 - may be totally conscious (Focal-aware) but unable to respond verbally because of aphasia/motor deficit. Pt may be aware during seizure, but seizure is occurring in the Broca’s area, may look like pt cannot follow commands or has alerted mental status
- Most common location is temporal lobe
 - Common in the uncus formation and the hippocampus → can maybe explain the amnesia a person may experience
- Most common motor manifestation is automatisms



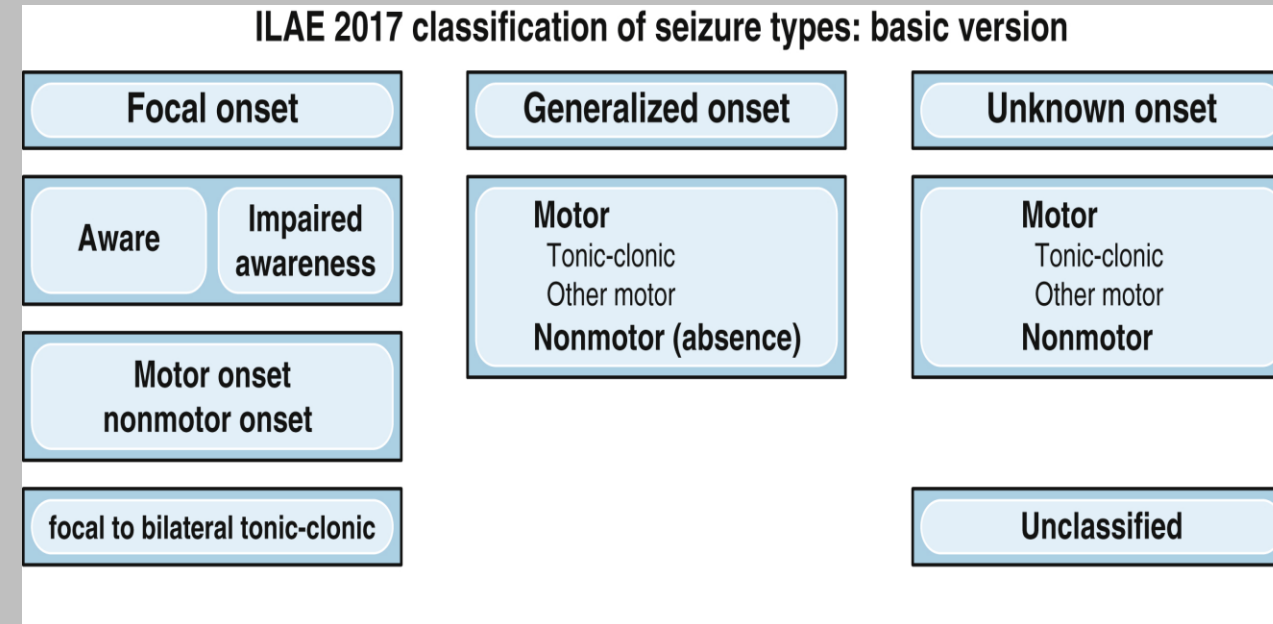
Focal-Aware Seizures: Differences in Origin



- Frontal and temporal lobe seizures can cause autonomic dysfunction
- Frontal lobe seizure can cause personality and executive function changes

Generalized Tonic-Clonic Seizure

- “*Grand mal*” is an old term
- **Do not have an aura** (such as a flash of light), but **may be preceded by a prodrome**, which is a vague sense that a seizure will occur; and this can occur days before the seizure happens. May include headache, mood changes, sleep disturbances or anxiety.
- Seizure onset is abrupt, most often with loss of consciousness and a generalized tonic contraction
- May be initiated with a series of myoclonic jerks
- Tonic, clonic and postictal phases --> discussed in next slide.



Generalized Tonic-Clonic Seizure

- **Tonic:** Upward eye deviation with eyes half open and the mouth open. There is involvement of the respiratory muscles usually produces a forced expiration that produces a loud guttural vocalization, often referred to as the epileptic cry.
- Cyanosis may occur during the tonic phase in association with apnea. The tonic phase gradually evolves to clonic activity.
- **Clonus:** Series of rhythmic jerky motions of clonus that slowly decrease in frequency over the course of the seizure
- **Postictal:** The individual is limp and unresponsive.
- Respiration is loud and snoring → as pt regains control over masseter and other jaw muscles. ABC's !!! Make sure airway is patent. Pt may awaken briefly with postictal confusion.
- Tongue biting common on the side of the tongue → not a reliable indicator if someone had a seizure.
- Incontinence of urine and stool.

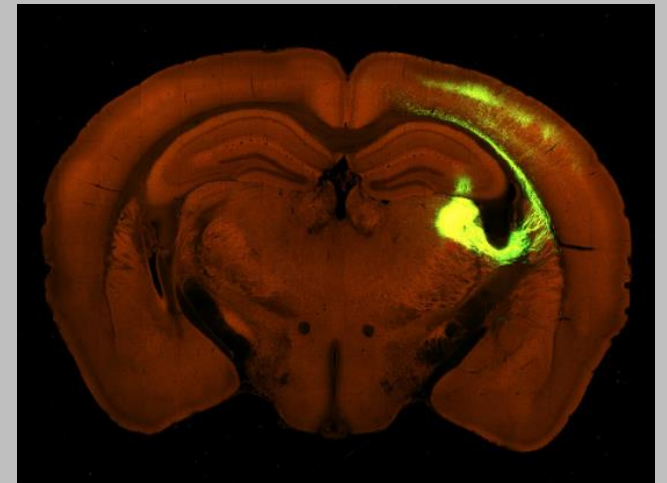
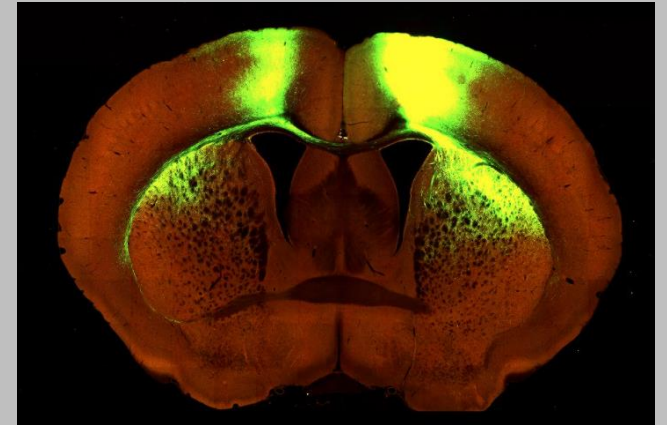
Generalized Seizures: Status Epilepticus

- **Neurological emergency that requires immediate intervention**
- Very clinically relevant concept. “Status” in the ED.
- Goal of therapy is to stop seizure activity before neuronal injury → Neuro-excitotoxicity, metabolic homeostasis is altered. Time is tissue survival, like stroke
- Definition: seizure that lasts more than a certain number of minutes or a series of seizures in rapid succession without full neurological recovery in the interictal period
- Typically, status epilepticus is considered **at 5 minutes for convulsive seizures** and **at 10 minutes for nonconvulsive seizures** → https://www.uptodate.com/contents/nonconvulsive-status-epilepticus-classification-clinical-features-and-diagnosis?search=status%20epilepticus%20definition&source=search_result&selectedTitle=4~150&usage_type=default&display_rank=4#H327107905
- Any seizure can evolve into status epilepticus
- First line treatment is lorazepam (Ativan) at 0.1 mg/kg/dose, up to 4mg max. (<https://pubmed.ncbi.nlm.nih.gov/36610189/>) → Dr. Goldstein lectures
- It is important to recognize that **most cases of status epilepticus do not have a history of epilepsy**

Generalized Seizures:

Focal to Bilateral Tonic-Clonic Seizures

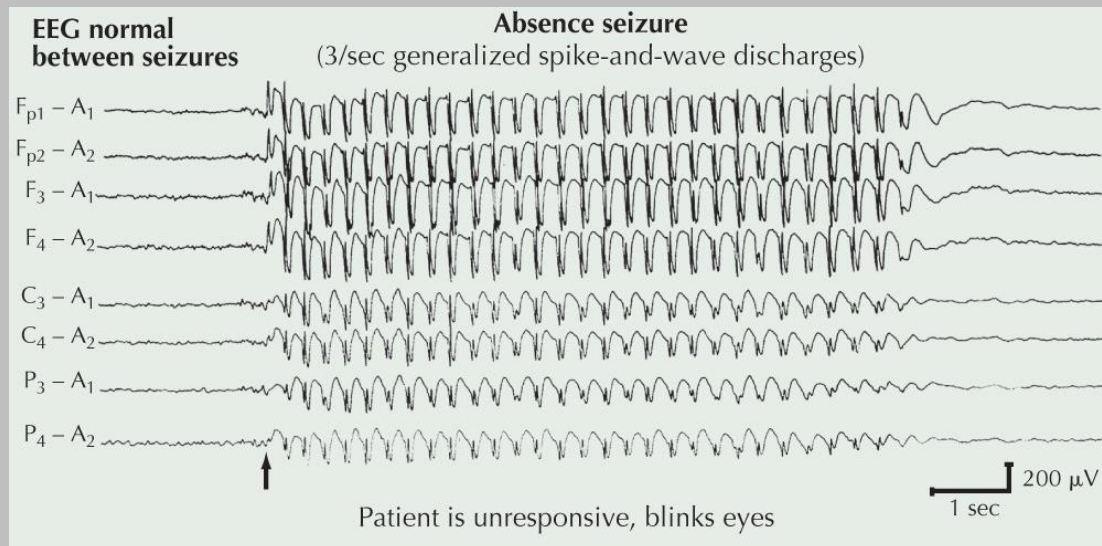
- A lot of generalized seizures can start as focal-aware or FIAS and progresses to a generalized seizure → due to hypersynchrony of neurons causing pathological action potential firing. I.e one neuron firing in an hyper-excitatory, aberrant pattern may cause resting membrane potential of another neuron to be closer to threshold (more likely to fire an AP!!)
- The pattern of evolution may be clonic-tonic-clonic in some instances.
- The bilateral tonic phase may be asymmetrical (e.g. flexion on one side and extension on the other), also known as figure-of-4



Generalized Seizures: Absence Seizures



- Sudden blank stare with motor arrest, usually lasting less than 15 seconds
- Usually unresponsive and unaware
- Returns immediately to a baseline level of function with no postictal confusion, important for step exams
- EEG during absence seizure: 3-5 hz spike & dome/wave
- Resemble slow-wave sleep EEG. Behavior during absence seizure resembles a sleep-like state.
- High-yield for boards
- EEG hallmark of a typical generalized absence seizure is generalized 3- to 5-Hz spike-and-wave activity with a normal interictal background.
- Hyperventilation is can be a trigger, can be used for diagnosis



- Clinical presentation may look something like this: A child presents to the pediatrician for complaints of "staring off into space" or "not paying attention or focusing in class or at home when told to do things". If absence seizure is suspected, the physician will ask the child to hyperventilate to see if this will elicit an absence seizure. This will help confirm our diagnoses and this pt will be most likely treated with ethosuximide.
- Treatment: **ethosuximide**, non -selective inhibition of Ca⁺ T-type channel Cav1, Cav2, Cav3 variants
- Long term prognosis: most outgrow

Generalized Seizures: Febrile Seizures

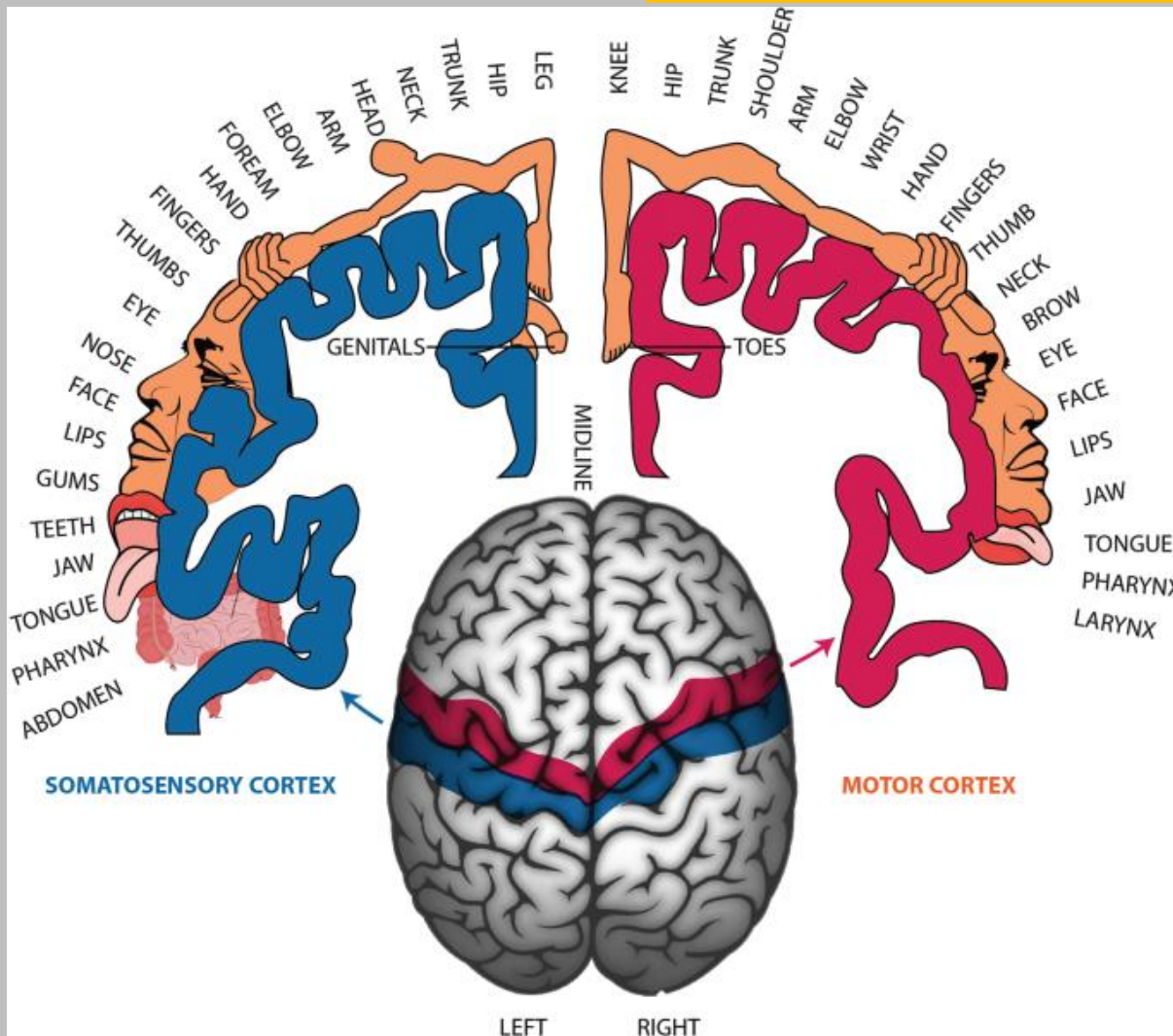
- Affects 2%–5% of all children, mostly between 3 months and 6 years
- Most common form of seizure in children
- Benign in themselves in the vast majority, but do increase the risk of other epilepsies later in life (7% by age 25)
- Most febrile seizures are generalized tonic-clonic in presentation.
- May be prolonged to qualify for the definition of febrile status epilepticus.
- Can cause brain injury (especially hippocampus)
- On step 2 boards, if pt has febrile seizure and you are in the ER, treat with antipyretic such as acetaminophen and discharge the pt once vitals are stable and follow up.
- There is evidence for strong genetic influence (likely polygenic)
- Some genetic epilepsy syndromes are known to start with febrile seizures (Especially Dravet syndrome)



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Classifying seizures... *the importance of behavioral characterization*

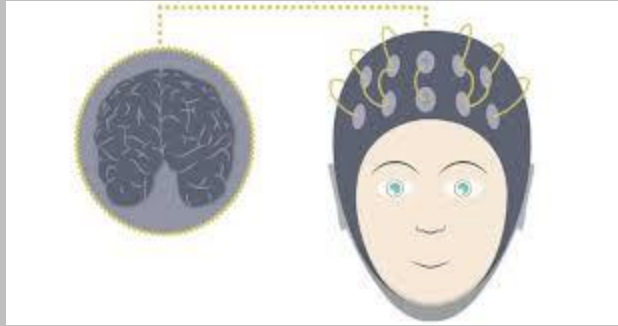


Jacksonian march (seizure)

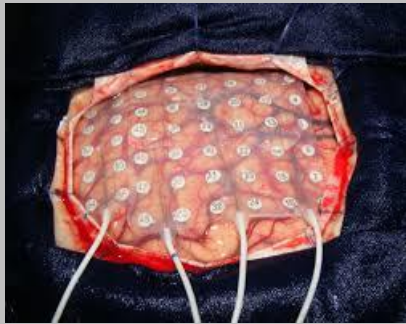
- One kind of a focal aware seizure, so consciousness is not lost or altered and EEG activity is on one hemisphere.
- Characteristic features of Jacksonian march are (1) it only occurs on one side of the body; (2) Seizure activity on EEG will essentially “March across the homunculus” be it somatosensory or motor.
- This will manifest as twitching or a tingling sensation in a limb or weakness in a finger, a big toe or the corner of the mouth, then **marches** over a few seconds to the entire hand, foot or facial muscles. This reflects the spread of abnormal neuronal activity to more parts of the brain.

EEG evaluation — gold standard for seizure detection, done after CT/MRI

Surface EEG



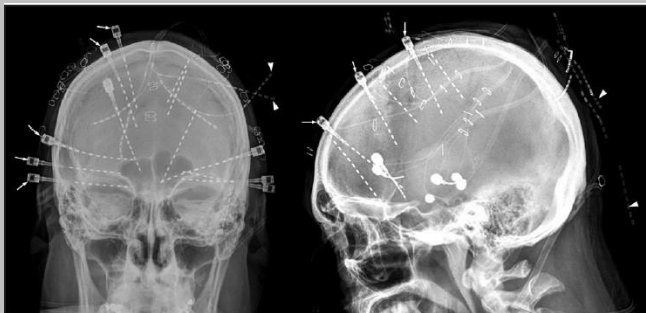
Epidural EEG



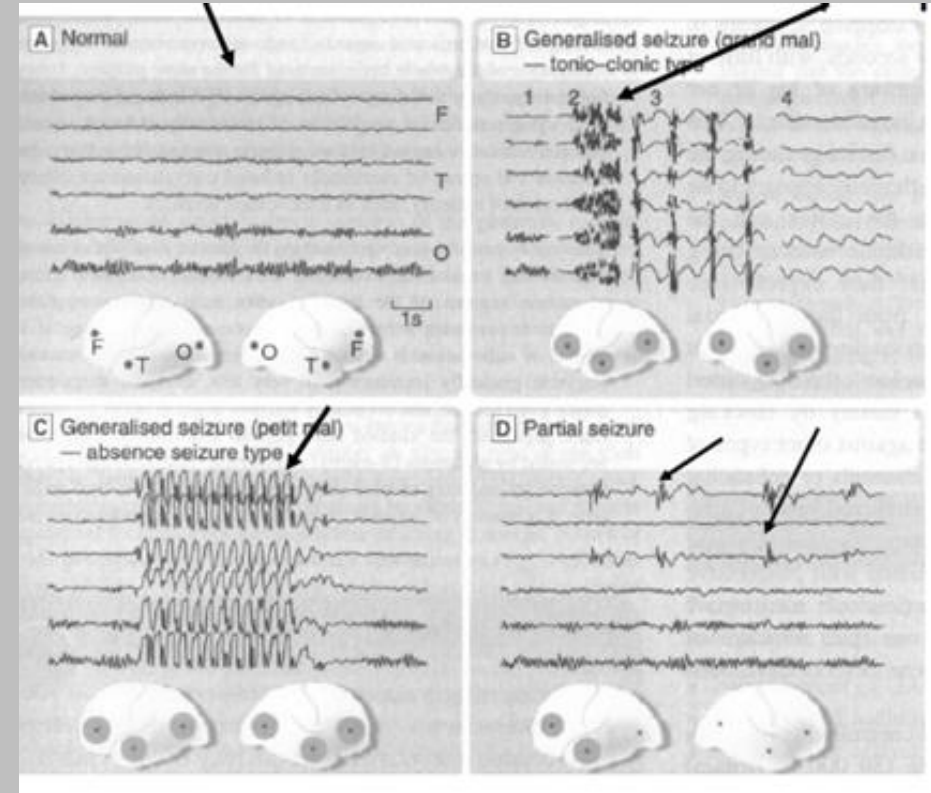
- More precision in localizing seizure focus, yet more invasive

- Craniotomy and surface electrodes on surface of brain

Intracranial electrode EEG



Spatial & temporal characterization of abnormal activity.



- CT= very quick, usually ordered before MRI in most emergency settings. CT will be ordered first in order to rule out any structural defects that may be causing the seizure. MRI is more expensive and can help identify more subtle lesions. MRI's take longer to complete than CT scans.

Pseudo-seizures:

- * Behavioral events that look like seizures but with no epileptiform activity on EEG.
- * Psych manifestation of some underlying issue.
- * Treat: **psychotherapy**, though seizures may recur during times of stress

What is epilepsy?

Epilepsy:

Disease or syndrome in which chronic, recurrent, unprovoked seizures is the salient feature.

Diagnosed by extensive clinical work-up

i.e. documenting frequency, duration, behavioral profile, etc of seizures,

EEG, MR Imaging, bloodwork, PMH, age, family history, genetics, diet, lifestyle, etc...

Disease of the brain with:

(1) at least two unprovoked seizures >24 hours apart

(2) One unprovoked or reflex seizure in an individual with a high risk of subsequent seizures (e.g., after traumatic brain injury, stroke, CNS infections)

(3) diagnosis of an epilepsy syndrome

- **Seizures and epilepsy are not just about neurons. Mutations can cause defects in astrocytes, endothelial cells and channels associated with these structures.**

***Think about tripartite synapse**

65 MILLION: Number of people around the world who have epilepsy.

3.4 MILLION: Number of people in the United States who have epilepsy.

1 IN 26 people in the United States will develop epilepsy at some point in their lifetime.

150,000: Number of new cases of epilepsy in the United States each year

6 OUT OF 10: Number of people with epilepsy where the cause is unknown.

- About 1 in 100 people in the U.S. has had a single unprovoked seizure or has been diagnosed with epilepsy
- **BETWEEN 4 AND 10 OUT OF 1,000:** Number of people on earth who live with active seizures at any one time.

Electro-clinical syndromes arranged by age at onset *

Neonatal period

Benign familial neonatal seizures (BFNS)
Early myoclonic encephalopathy (EME)
Ohtahara syndrome

Infancy

Migrating partial seizures of infancy
West syndrome
Myoclonic epilepsy in infancy (MEI)
Benign infantile seizures
Benign familial infantile seizures
Dravet syndrome
Myoclonic encephalopathy in nonprogressive disorders

Childhood

Febrile seizures plus (FS+) (can start in infancy)
Early onset benign childhood occipital epilepsy (Panayiotopoulos type)
Epilepsy with myoclonic atonic (previously astatic) seizures
Benign epilepsy with centrotemporal spikes (BECTS)
Autosomal-dominant nocturnal frontal lobe epilepsy (ADNFLE)
Late onset childhood occipital epilepsy (Gastaut type)
Epilepsy with myoclonic absences
Lennox-Gastaut syndrome
Epileptic encephalopathy with continuous spike-and-wave during sleep (CSWS)
 including: Landau-Kleffner syndrome (LKS)
Childhood absence epilepsy (CAE)

Adolescence - Adult

Juvenile absence epilepsy (JAE)
Juvenile myoclonic epilepsy (JME)
Epilepsy with generalized tonic-clonic seizures alone
Progressive myoclonus epilepsies (PME)
Autosomal dominant partial epilepsy with auditory features (ADPEAF)
Other familial temporal lobe epilepsies

Incomplete List of Epilepsy Syndrome

- Don't memorize this
- All effect ADL and quality of life for patients and family.
- Increased mortality due to a number of factors.
- All these syndromes have salient features that help distinguish one condition from another differentiate each
- Effects all ages, diverse clinical syndromes that separate each syndrome from the other
- Age of onset matters a lot, adolescent vs neonatal, etc
- Epilepsy fellow will be familiar with these syndromes
- More epilepsy syndromes to come in later slides

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Causes of epilepsy

These categories are becoming blurred due to increased use of genomic sequencing in clinical medicine.

1. Structural/metabolic: (MRI/CT positive)

A distinct condition or disease that has been demonstrated to be associated with a substantially increased risk of developing epilepsy.

- Neuronal migration, brain malformation ex later. Visible on MRI

2. Genetic: (MRI/CT Negative)

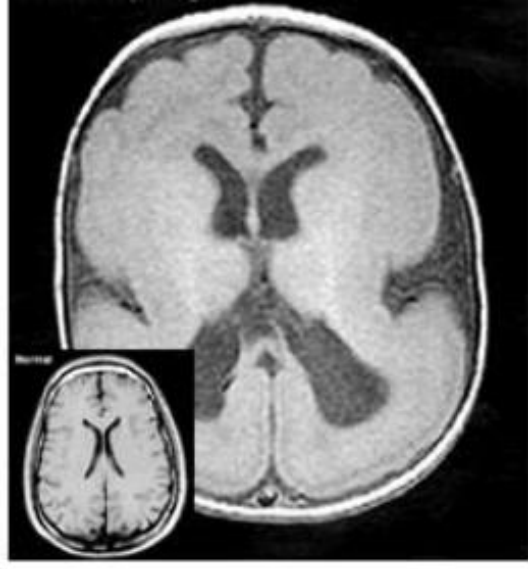
a direct result of a known or presumed genetic defect(s) in which seizures are the core symptom of the disorder.

- **Genetic mutations** encoding for Ach, GABA, Glutamate receptors, Na⁺, K⁺, Cl⁻ ion channel. **Not generally visible on MRI**

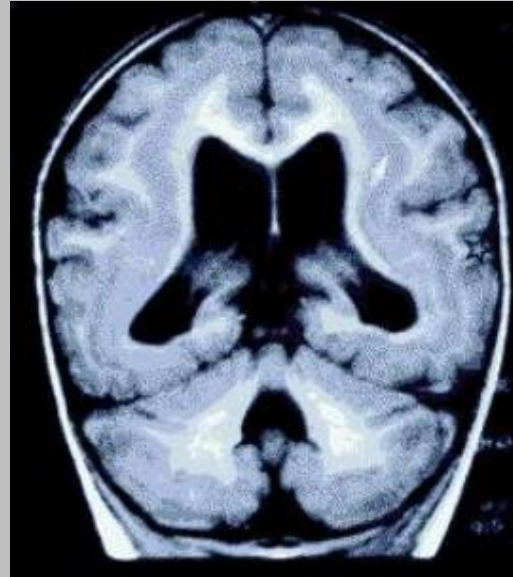
3. Unknown/idiopathic

Neocortical malformations are associated with epilepsy

Lissencephaly- no sulci/ gyri
Smooth

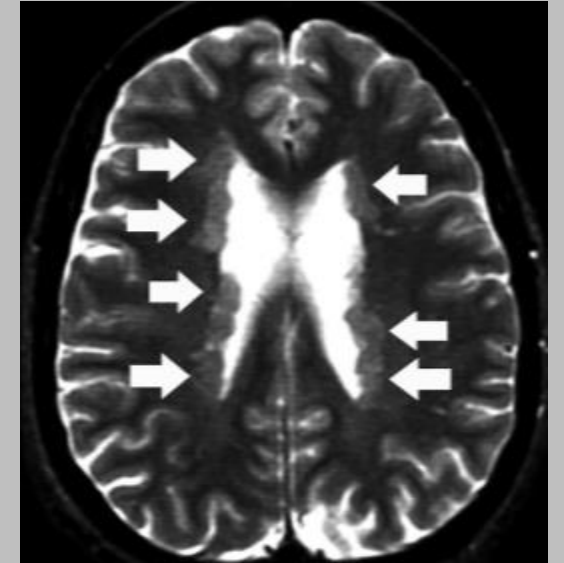


Double-cortex syndrome

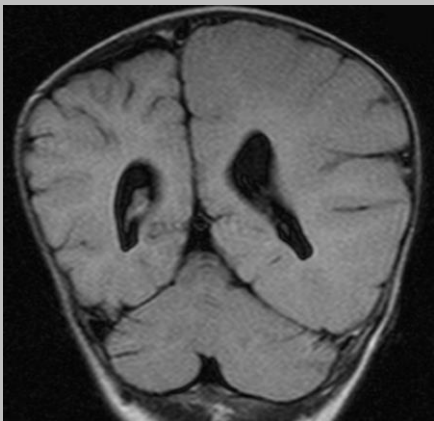


Correction:
Neurons
caught in
subcortical
white
matter in
DCX!!

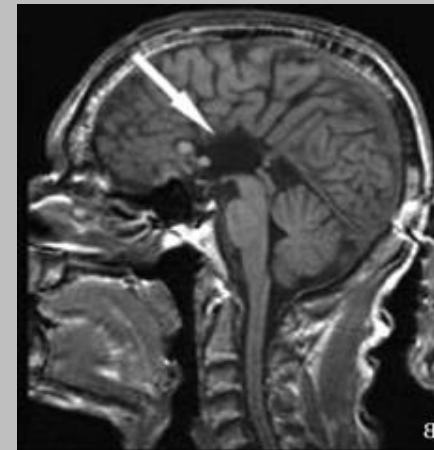
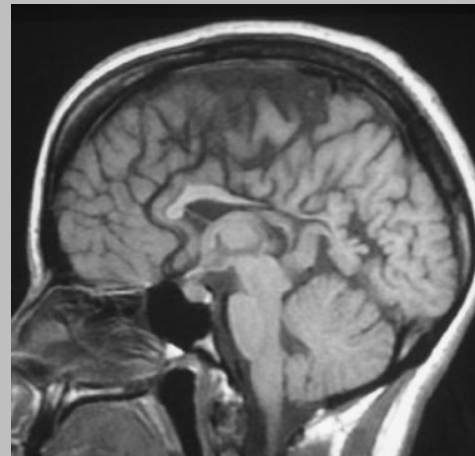
Periventricular nodular heterotopia



Hemi-meganecephaly



Callosal dysgenesis or agenesis



- Different issue compared to people who get corpus callosotomies to treat seizures

Structural/Metabolic causes of both seizures and epilepsy

- Malformations of neocortical development
- Neurocutaneous syndromes (Tuberous sclerosis, Sturge-Weber syndrome)
- Brain Tumor
- Infection (meningitis, AIDS, neuroinflammation)
- Trauma Head injury (25% acute seizures; 10-25% chronic epilepsy)
- Brain Angioma/ vascular malformation
- Peri-natal insult (fetal anoxia)
- Stroke (11% of all new cases of adult-onset epilepsy)

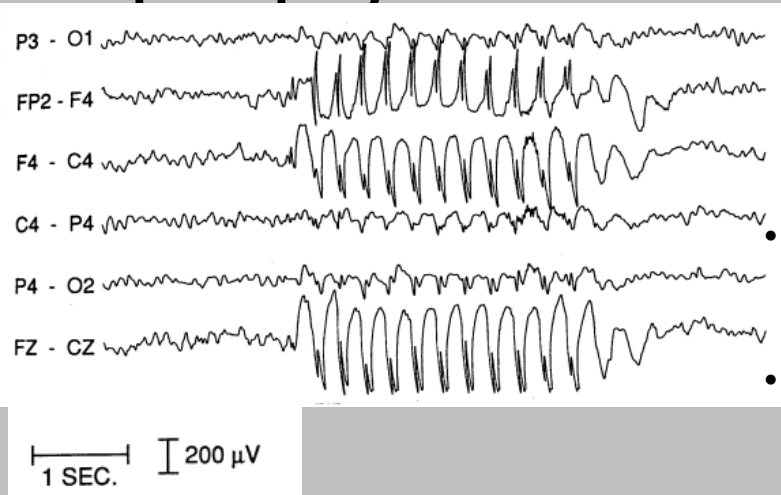
Note: In next slides we will go over a lot of different types of epilepsy and seizures. For your studying and your exam, look for **SALIENT and Distinguishing features** in the descriptions of each disorder. This will help you distinguish one disorder from another.

- Ex) genes, patient presentation, inheritance, age of patient, etc... --> tying these concepts together for each disorder is what Dr. Ramos and myself will test you on

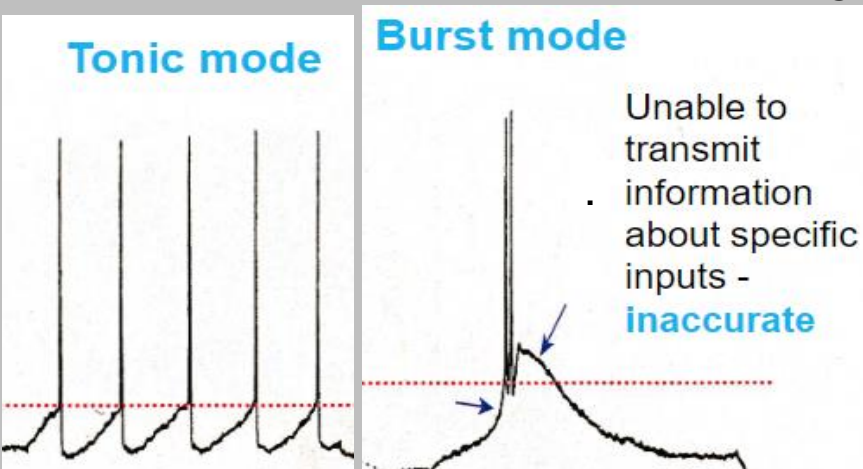
Selected Epilepsy Syndromes (Testable on Exam): Dravet Syndrome

- De novo mutation in **SCN1A gene** - encoding the α -1 sodium channel subunit of voltage-gated sodium channel in neurons.
- **SCN1A gene mutation** has been shown to make Nav 1.1 Na⁺ channel in neurons open more easily, open for longer periods of time, or re-open faster after closing. More AP's fired!!
- **Good example of epilepsy syndrome with known gene and direct mechanism related to membrane physiology of neurons (action potentials)**
- MRI Positive or **Negative?**
- Also called **severe myoclonic epilepsy of infancy**
- Typical clinical presentation: previously normally developing infant has febrile status epilepticus at 6 months old, then get recurrent generalized seizures often triggered by fever.
- After 1 year of age, other seizure types appear, including myoclonic seizures, absence seizures, and focal-impaired-awareness and atonic seizures
- The prognosis is poor (intellectual disability, ataxia and spasticity)

Cacna1H/ Cacna1A mutations & Absence Epilepsy



- Thalamus has Thalamocortical relay cells (TCR neurons) which are involved in the regulation of sleep-wake states as well as transmission of sensory inputs to the neocortex for conscious appraisal. **When TCR neuron are hyperpolarized (more negative charge inside cell) this cause de-inactivation of Type T Ca²⁺ channels and eventual creation of burst firing patterns seen in absence epilepsy and SWS.** Gates incoming sensory info to cortex. Tonic mode occurs in (Ach) depolarized TCR neurons (awake and rem sleep)
- TCR neurons have 2 modes, Tonic and burst. Tonic mode occurs in depolarized TCR neurons and this permits consistent signal transmission along TCR neurons. Occurs in REM and wakefulness.
- Excessive calcium influx through the Type T Ca²⁺ channels in a hyperpolarized TCR neuron is the mechanism from burst mode. This locks the TCR neurons into the characteristic 3 hz spike and dome patterns. This oscillation pattern inhibits transmission of external environmental stimuli.
- The **Cacna1H** mutation thalamic Type T Ca²⁺ channels of Thalamocortical relay cells cause aberrant calcium influx into TCR neurons during wakefulness of patients with absence epilepsy, causing the EEG finding shown.
- Cacna1a gene may also be implicated in disease process.



- Treatment: **ethosuximide**, non - selective inhibition of Ca²⁺ T-type channel Cav1, Cav2, Cav3 variants
- Long term prognosis: most outgrow

Lennox Gastaut syndrome

- Lennox-Gastaut Syndrome (LGS) is a lifelong, developmental epileptic encephalopathy
- Onset before age 8, with peak onset between the ages of 3 and 5.
- A recent study estimates the incidence of LGS to be between 14.5 and 28.0 per 100,000 individuals, with a prevalence ranging from 5.8 to 60.8 per 100,000 individuals*
- **Characterized by frequent seizures, including atonic, tonic, and atypical absence seizures → creates neuropsychiatric issues such behavioral disorders and intellectual disabilities.**
- LGS can arise from genetic mutations, cortical malformations, brain tumors, neurocutaneous disorders such as tuberous sclerosis complex, hypoxic-ischemic injury, meningitis, or head trauma. No singular cause. However, in approximately 40% of cases, no identifiable cause is found
- EEG findings may include **interictal 2.5 Hz slow spike-wave discharges and generalized paroxysmal fast activity during sleep.**
- Gene to know: **GABRB3 mutation**, implicated in LGS pathophysiology, makes beta-3 subunit of GABA-A receptor
- Diagnosis relies on clinical presentation, EEG findings, brain magnetic resonance imaging, and genetic testing.
- LGS significantly impacts patients' quality of life, placing considerable strain on caregivers and healthcare systems. Multidisciplinary care needed.

Landau-Kleffner Syndrome

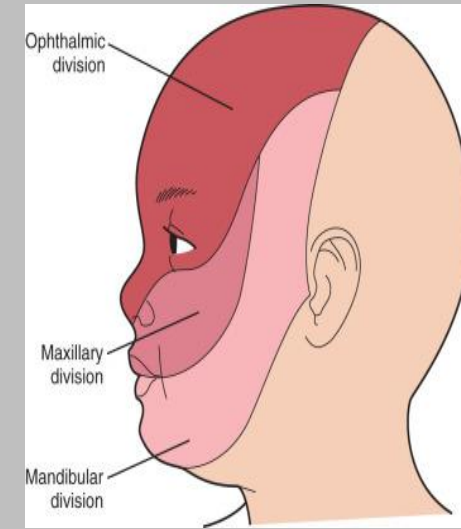
- Landau-Kleffner syndrome (LKS) is a rare age-related epileptic encephalopathy of children **aged roughly 3 to 8 years old with previously normal neuropsychiatric development.**
- Disorder is characterized by **language regression** and seizures and is often associated with social cognitive deficits.
- Disorder characterized by a **developmental regression Broca's and Wernicke's areas, and electroencephalogram (EEG) anomalies located in those areas.**
- **New onset receptive and then later expressive aphasia in previously normally developing individual is hallmark of this disease**
- The seizures, if present, are mainly absence seizures or tonic-clonic episodes which may occur more often during sleep.
- Disorder associated with the **GRIN2A gene**
- This gene makes GluN2A protein: This protein is a subunit of NMDA receptors in neurons.
- NMDA receptors: These are crucial glutamate-gated ion channels in the brain that control the flow of sodium, exciting neurons and transmitting signals.
- Role in the brain: NMDA receptors are known to be involved in learning, memory brain development, and synaptic plasticity.

Autosomal dominant partial epilepsy with auditory features (ADPEAF)

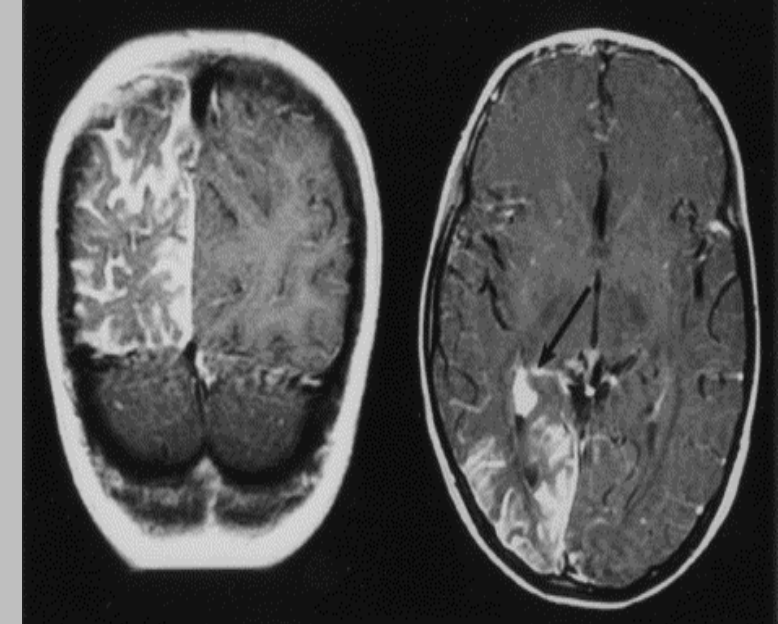
- Autosomal dominant epilepsy with auditory features (ADEAF) is a focal epilepsy syndrome
- During the seizure, **a patient may have auditory symptoms i.e humming, buzzing, or ringing and/or receptive aphasia**
- The receptive (Wernicke's) aphasia during the seizure consists of a sudden onset of inability to understand spoken or written language without general confusion.
- Other less common symptoms during the seizure may include visual, olfactory, vertiginous, motor, and autonomic symptoms.
- Age at onset is usually in adolescence or early adulthood (age 10-30 years). The clinical course of ADEAF is benign. Seizures are usually well controlled after initiation of medical therapy.
- Since this disorder has an **AUTOSOMAL DOMINANT** inheritance. Use **Family History** to potentially help you make the diagnosis. If similar presentations have occurred in the patient's family before, this may help you in making your differential diagnosis. Rates of de novo pathogenetic mutations are very low
- Mandatory **criteria to make diagnosis** include a **normal MRI**, as well as **Focal sensory auditory seizures and/or focal cognitive seizures with receptive aphasia without generalized confusion during the seizure.**
- Genes to know that are implicated in this disorder include *LGII*, *MICAL1*, or **RELN**.
- Know **RELN** for test → codes for Reelin

Sturge-Weber Syndrome

- Nice example of a disorder with a known gene, robust physical/clinical presentation, characteristic MRI signature
- Congenital nonhereditary anomaly of neural crest derivatives.
Somatic mosaicism of an activating mutation in one copy of the GNAQ gene. **GNAQ gene vital for blood vessel development
- Somatic Mosaicism: DNA mutations or alternations in copy number of cellular genomes occur post-fertilization that create two distinct populations of cells. Somatic mosaicism mutations are not inherited, but genetic mosaicism mutations can be.
- Capillary vascular malformations → port- wine stain in Cranial N. V1/V2 distribution; risk of neuro-ocular issues
- ipsilateral leptomeningeal angioma with calcifications → seizures/ epilepsy; intellectual disability; early-onset glaucoma.

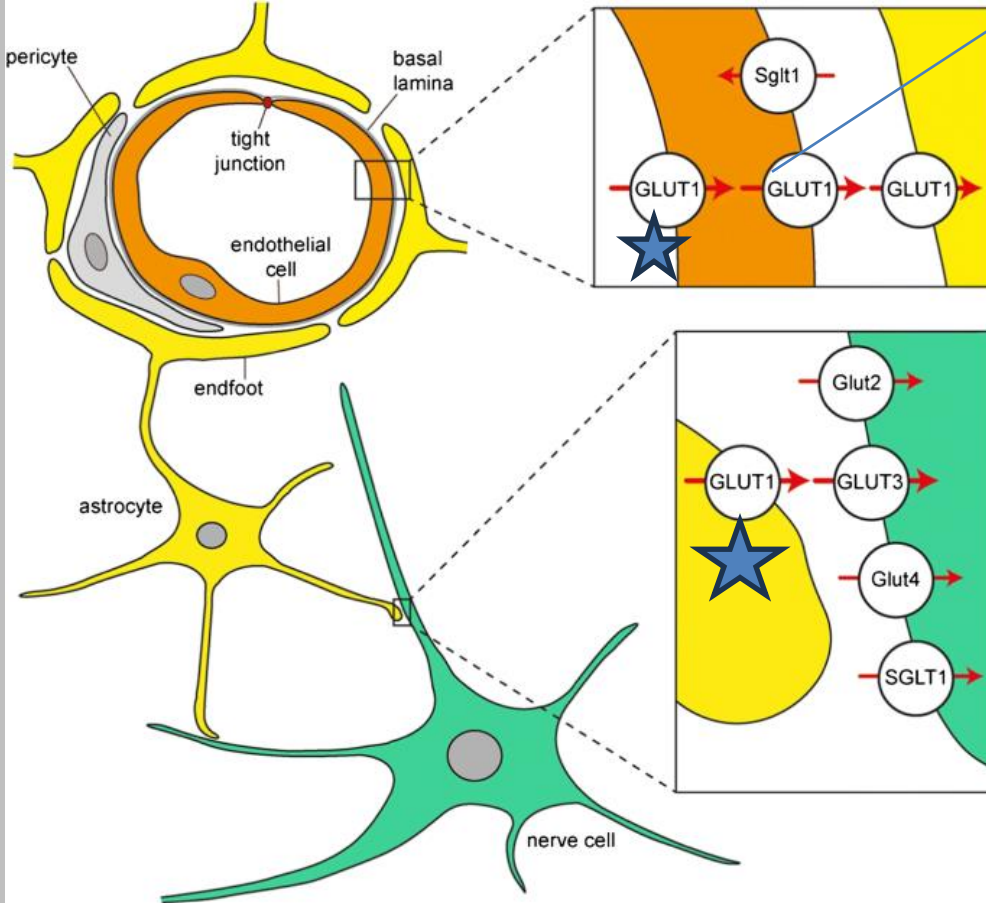


Port wine stain



Endothelial cells & epilepsy – Glut1 gene (glucose transporter) → GLUT1 deficiency syndrome

- Astrocytes get glucose from cerebral arteries. Astrocytes transform glucose to lactate and help shuttle lactate to neurons for ATP production



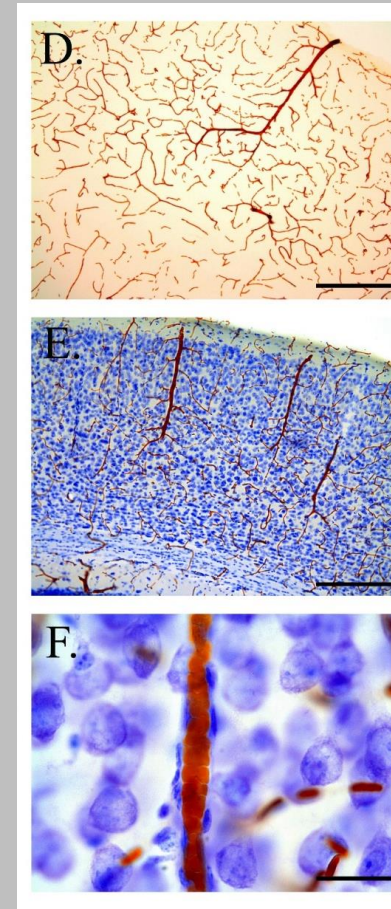
- Trouble transporting glucose from cerebral arteries into astrocytes

Ex) after ruling out infectious agent/meningitis and the patient has had a seizure and you see low glucose in the CSF, think about this disorder as a potential cause of the seizure

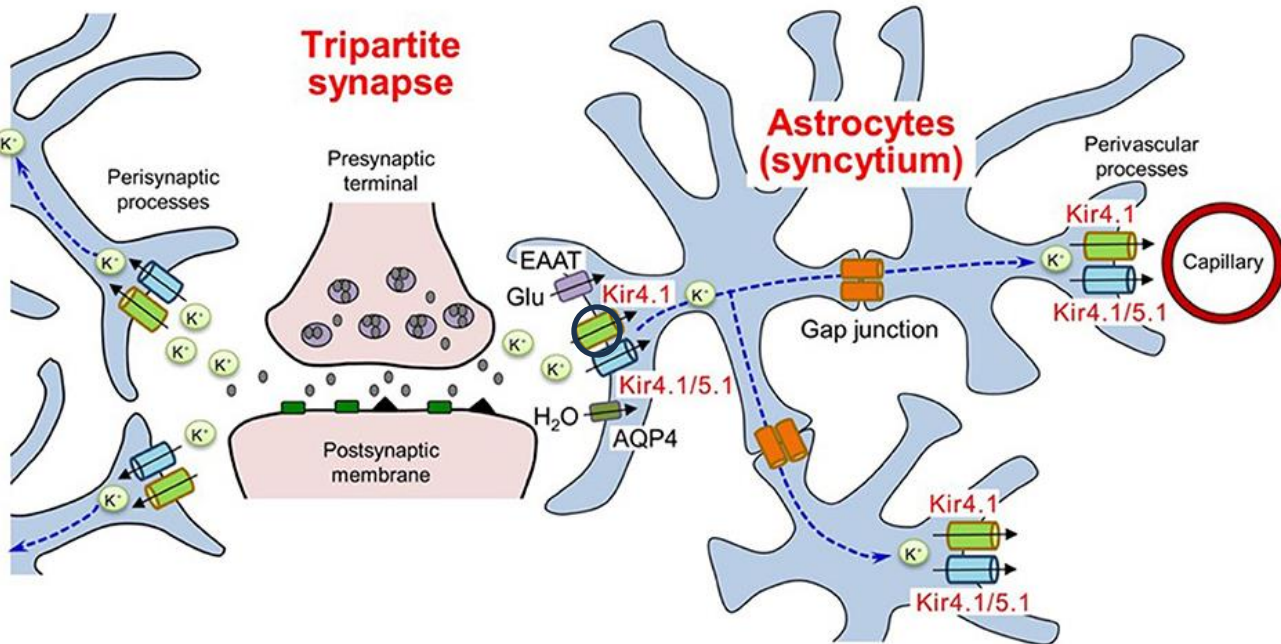
- Glut 1 in brain. Glut 4 mostly in skeletal muscle and adipose

GLUT1 deficiency syndrome

- symptoms can include developmental delay, intellectual disability, movement problems, and frequent seizures (diverse types of seizures like atonic, tonic-clonic, etc).
- Problem in vascular transport out of blood into brain
- **Hypoglycorrhachia: low concentration of glucose in the CSF**
- Think of BGL as your 6th vital as a clinician!!!



Astrocytes & Epilepsy: EAST Syndrome



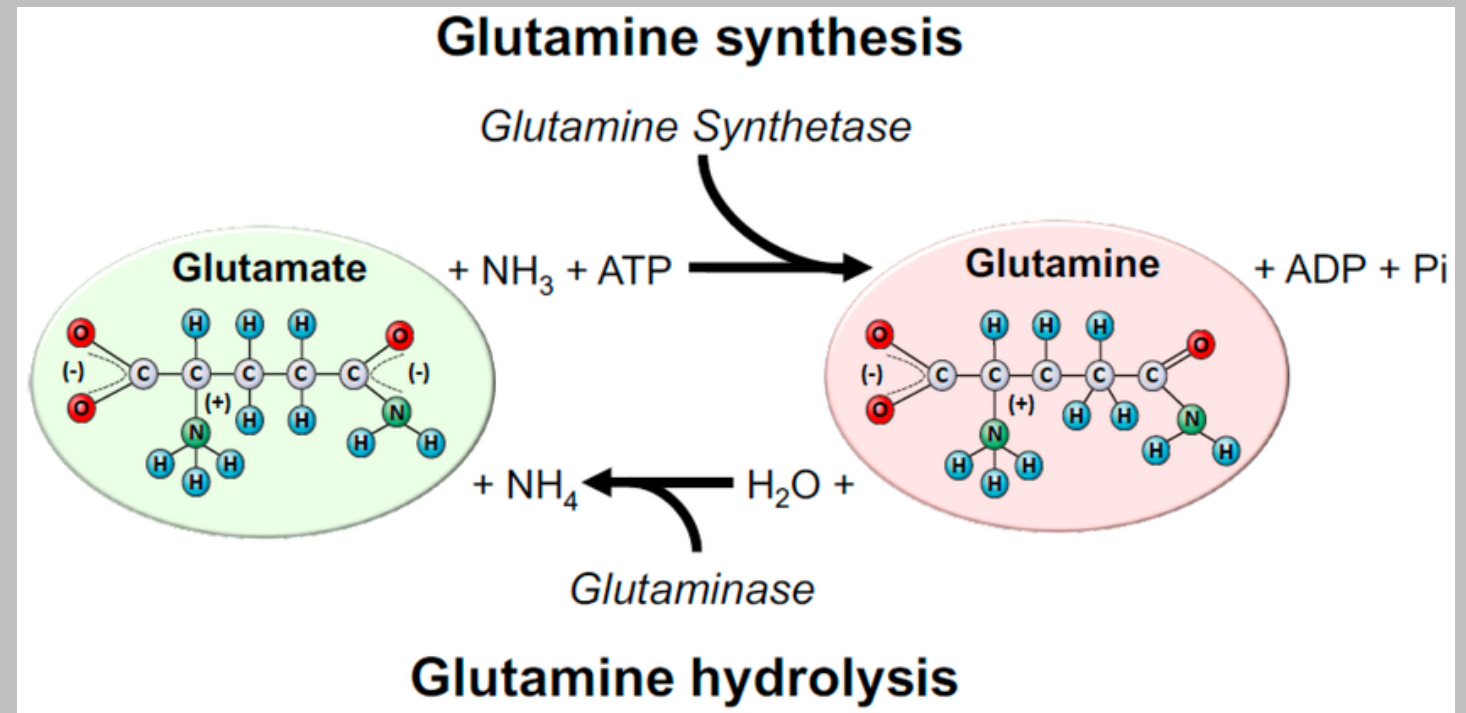
EAST syndrome: *KCNJ10* gene → Epilepsy, ataxia, sensorineural deafness and kidney tubulopathy.

- *KCNJ10* gene encodes an ATP-sensitive K⁺ channel found only in astrocytes (*Kir4.1*) → encodes channel for astrocyte to clear out extracellular potassium.
- Too much potassium in extracellular space results in depolarization of neurons – hyperexcitability. Remember the massive role of K leak channel in RPM maintenance

Astrocytes & Epilepsy: Glutamine deficiency syndrome

Glutamine deficiency syndrome:

- Mutation of **Glul gene**, encodes **glutamine synthetase**, preferentially expressed in **astrocytes**.
- Onset of seizures in the first days or weeks of life. Affected infants show severely impaired psychomotor development with hypotonia, spasticity, lack of speech, poor visual fixation, feeding difficulties may necessitating tube feeding.
- **Increased ammonia may contribute to developmental issues. GABA/glutamate balance is disrupted**
- Glutamine hit with enzyme glutaminase → glutamate → GABA + Co₂ via glutamate decarboxylase (in neurons)

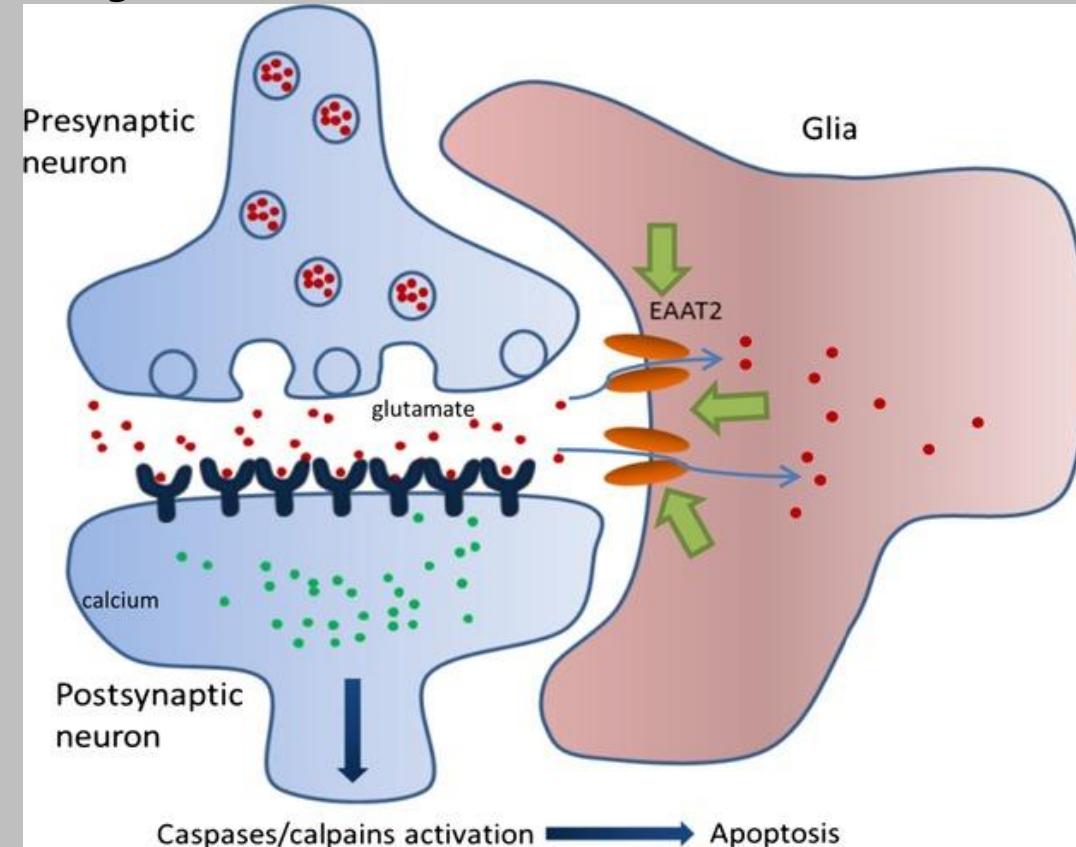


https://www.researchgate.net/figure/Glutamine-synthesis-and-hydrolysis-Glutamine-is-mainly-synthesized-by-the-enzyme_fig1_328480694

DEE- *Developmental and epileptic encephalopathy 41*

DEE41 syndrome:

- Mutation of **EAAT2 (Excitatory amino acid transporter)** gene → too much glutamate in extracellular space results in depolarization of neurons – hyperexcitability. Characterized by the onset of seizures in the first days or weeks of life. Affected infants show severely impaired psychomotor development with hypotonia, spasticity, lack of speech, poor visual fixation, and feeding difficulties sometimes necessitating tube feeding



Tuberous Sclerosis

Ashleaf spot



Facial angiofibroma

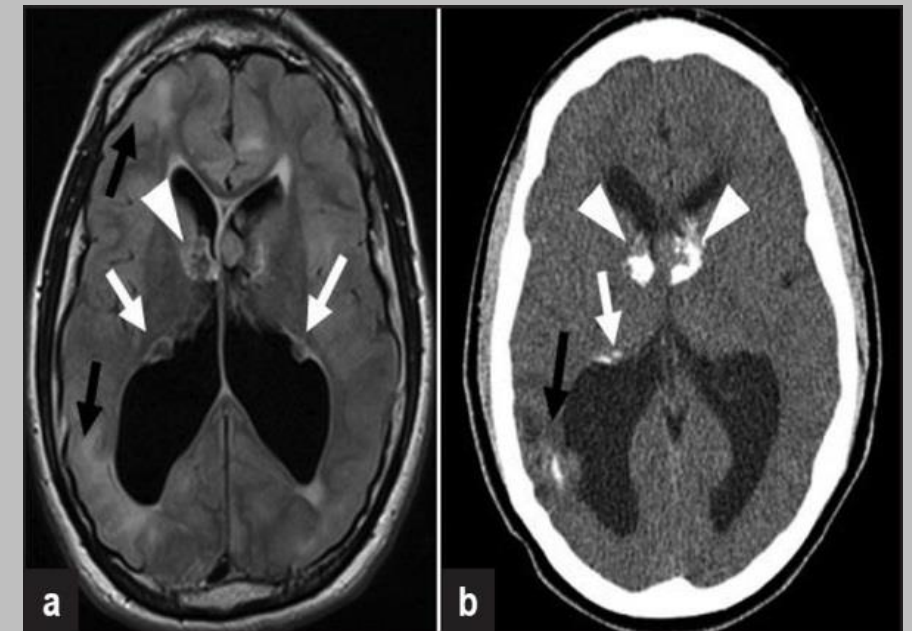


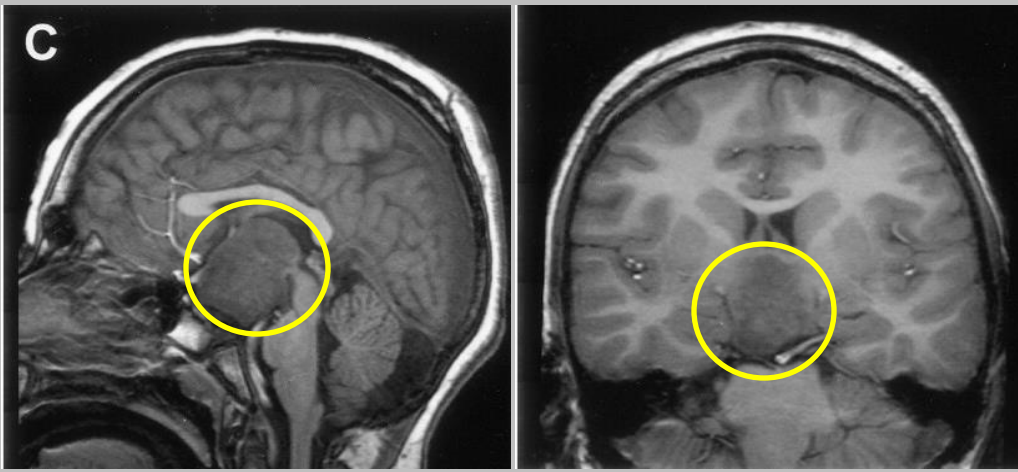
- Neurocutaneous disorder with **autosomal dominant inheritance pattern** (genes identified in 1990's)
 - **TSC1** chrom 9q → **hamartin**: tumor suppressor (downregulates mTOR)
 - **TSC2** chrom 16p → **tuberlin**: tumor suppressor (downregulates mTOR)
 - Genes involved may explain hamartoma presence in various systems
- Clinical mnemonic (**ASHLEAF**)
 - Ashleaf spots
 - Shagreen patches (skin hamartoma)
 - Heart rhabdomyosarcoma
 - Lung hamartomas
 - Epilepsy (brain hamartomas)
 - difficult to control → **can cause seizures**
 - Angiomyolipoma in kidney
 - Facial angiofibroma

Shagreen patch



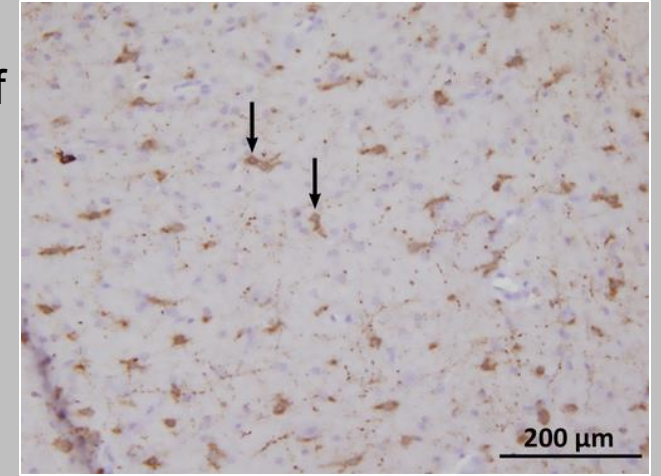
Brain Hamartomas





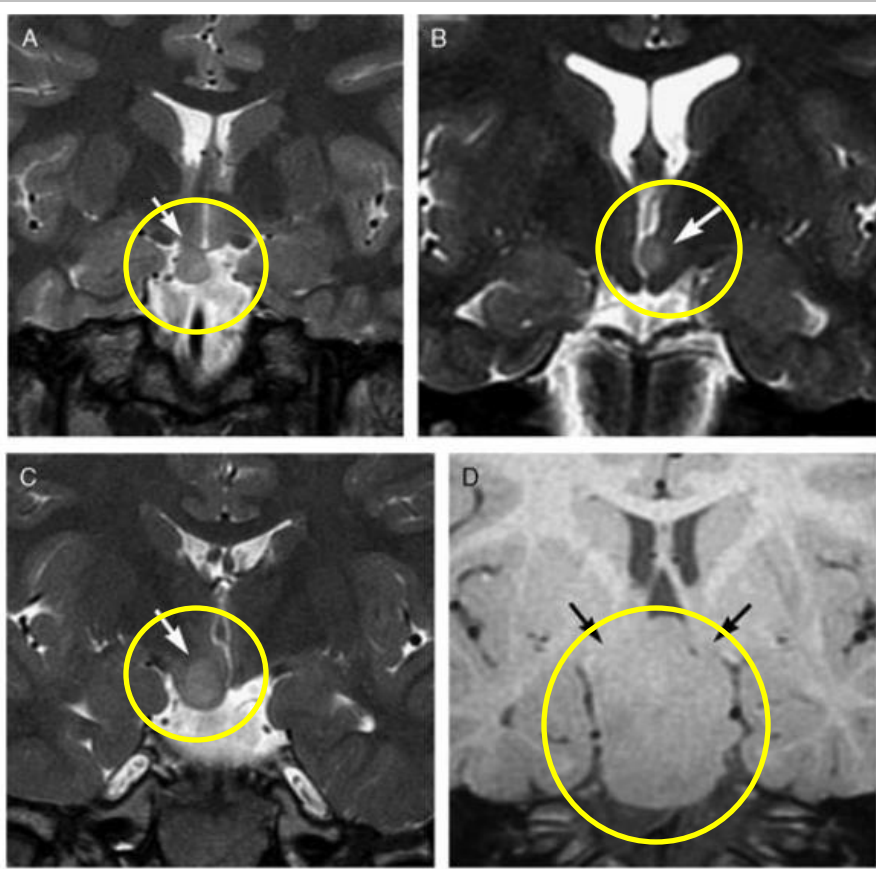
Hypothalamic hamartoma & seizures

- Brain hamartoma made up of neurons and glia. Not a tumor, just abnormal proliferation of tissue type in the area



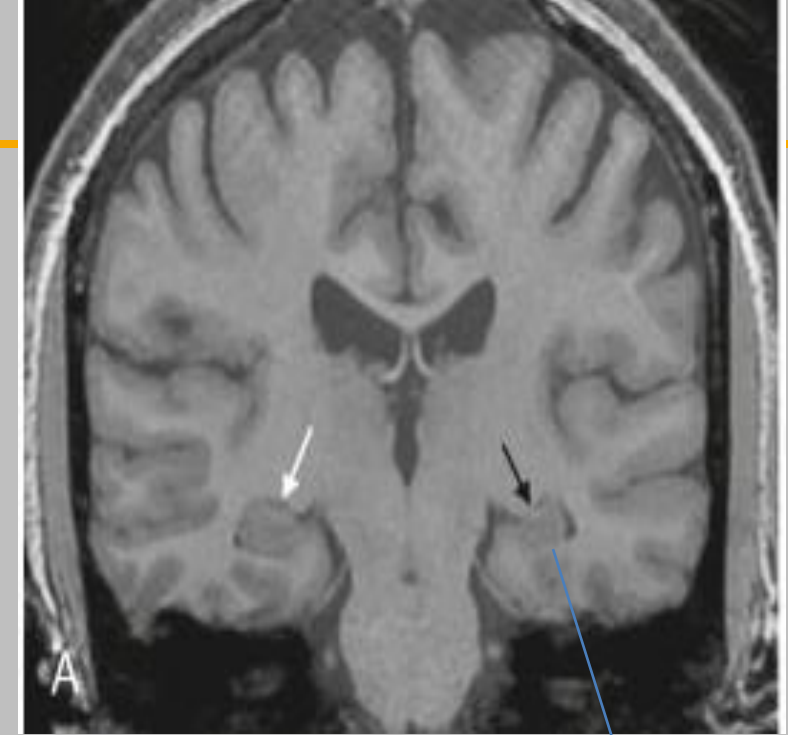
- Gelastic** seizures bouts of laughing.
- Dacrystic** seizures bouts of crying.
- Child may experience early puberty and irritability and aggression between seizures due to premature GnRH elevations and pulsations
- Treatment includes surgical resection.

JL Freeman et al 2004

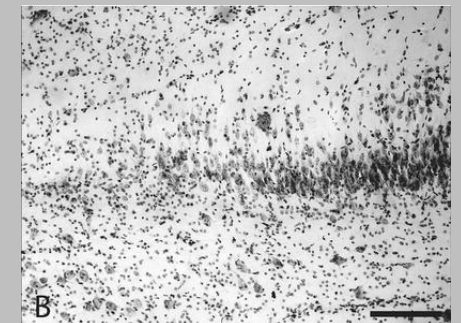
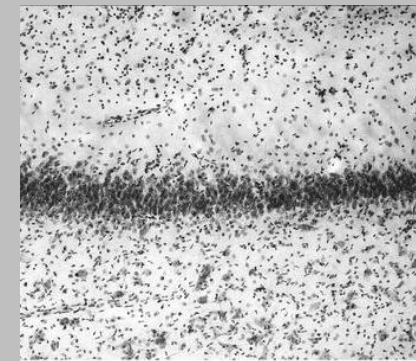


Medial Temporal Lobe Epilepsy

- About 30% cases are intractable. Minimal Pharmacological treatment
- Temporal lobe epilepsy accounts for 70% of intractable epilepsy
- MTLE usually manifest as Generalized Tonic-clonic seizure. Starts most commonly in hippocampus (MTL)
- MTL Sclerosis = **scarring** and **volume loss (atrophy)** of medial temporal structures (most notably **hippocampus**)
- Dx: **MRI**
- Characterized by neuronal loss and gliosis of the hippocampus and adjacent structures
- Unclear Pathophysiology of MTS

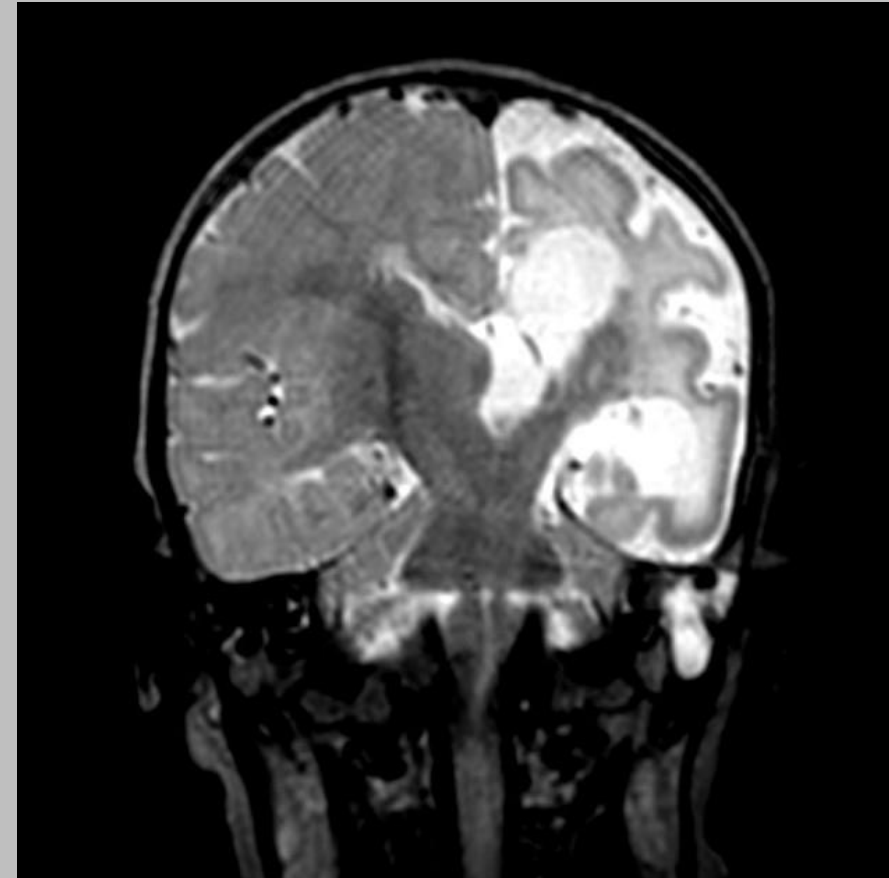


Sclerosis in
this
hippocampus



Autoimmune Encephalopathies: Rasmussen Syndrome (RAS-muss-en)

- Rasmussen Syndrome is a rare pediatric encephalitis that effects **one hemisphere of the brain**.
- 1.8 to 2.4 out of every 10 million people annually, or 0.18 per 100,000 people, usually in previously normally developing pediatric boys (aged 6-8 most common)
- Drug resistant seizures are a hallmark feature
- T-cell-dominated encephalitis with **activated microglial cells and reactive astrogliosis is apparent on histopathological examination**. What is causing this microglia activation and subsequent astrogliosis is unknown. No antigen that CNS immune cells attack has been identified
- Characterized by recurrent seizures, delayed developmental milestones, progressive cognitive decline. This can be acquired via through medical history.
- neuroimaging with chronic inflammatory changes, and progressive hemispheric atrophy
- In short, we really don't know what causes this disease
- Treatment involves a hemispherectomy of the affected brain lobe/hemisphere

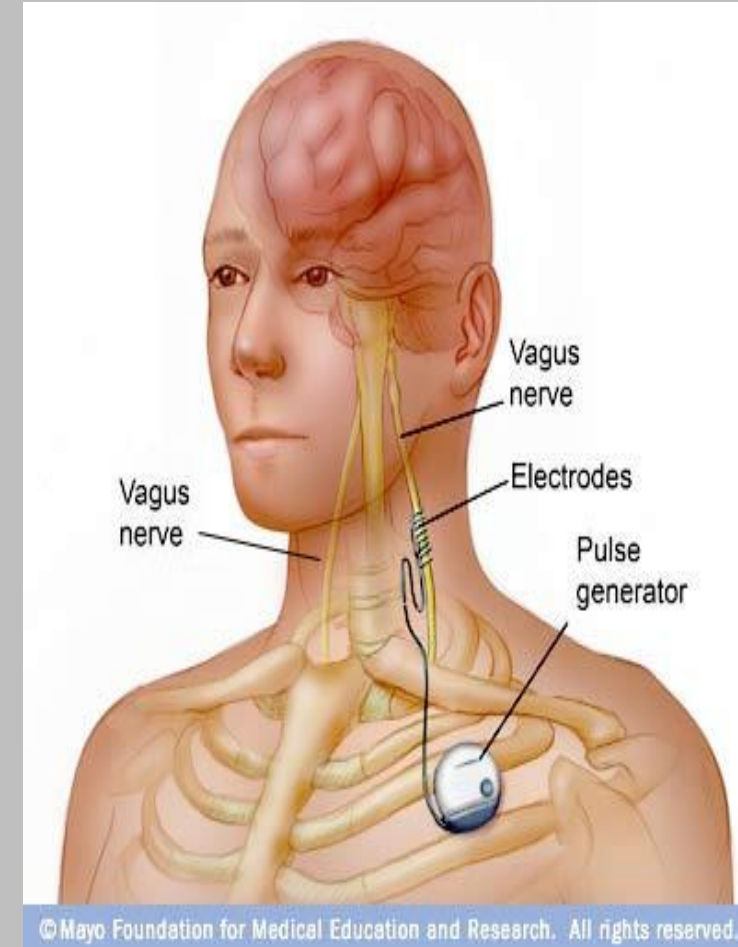


Autoimmune Encephalopathies: NMDA Encephalitis

- Anti-NMDA Receptor encephalitis is the most common (one affected out of 1.5 million people per year) autoimmune encephalitis
- **IgG antibodies against the NR1 subunit of the NMDA receptors in the central nervous system(CNS)** are a hallmark of this disease. Unlike Rasmussens, we have an idea about what antibody is attacking what antigen. These antibodies are identifiable in the serum or the cerebrospinal fluid (CSF)
- Plasma exchange helps reduce amount of Anti-NMDA receptor antibodies in serum
- Seems to be triggered by viruses such as Herpes simplex or ovarian teratomas (tumor)
- **Transvaginal ultrasonography** is the most crucial test for young women presenting with the illness due to the high incidence of underlying **ovarian teratoma**.
- As with most autoimmune disorders, women aged 25-35 are most commonly affected
- 5 clinical Stages: prodromal phase, psychotic phase, unresponsive phase, hyperkinetic phase, and recovery phase.
 1. Prodrome: can mimics common viral infections.
 2. Psychotic: Psych symptoms in adults. Includes visual or auditory hallucinations, acute schizoaffective episodes, depression, mania, apathy, anxiety, fluctuating sensorium, bizarre behaviors, hypersexuality, aphasia, amnesia, sleep-wake cycle disruption or eating disorders can appear rapidly within days to weeks in these patients with no prior psychiatric diagnosis. Children who are affected suffer less psych symptoms and more from movement disorders and seizures in this period
 3. Unresponsive phase: unresponsive phase characterized by mutism, decreased motor activity, and catatonia
 4. Hyperkinetic phase: Lip and tongue smacking common, as well as autonomic instability
 5. Recovery of language function and behavioral symptoms occurs at the end.

Treatments

- Goal of Therapy:
 - Ex: seizure freedom! >> reduction in seizure frequency, no complex seizures, simple partial seizures only
- Pharmacological → future Dr. Goldstein lecture
- Surgery
- Brain stimulation
- Vagus Nerve Stimulator – increase parasympathetic tone
- Ketogenic Diet – not entirely understood mechanism. May effect cellular metabolism enough to increase seizure threshold. Usually diet minimal in carbohydrates, this stimulates lipolysis ketone release from liver (liver cannot use ketones as energy) which may effect resting neuronal membrane potential and overall proclivity to fire action potential
- Stress Modulators
- Yoga
- Acupuncture
- Seizure Alert Dogs

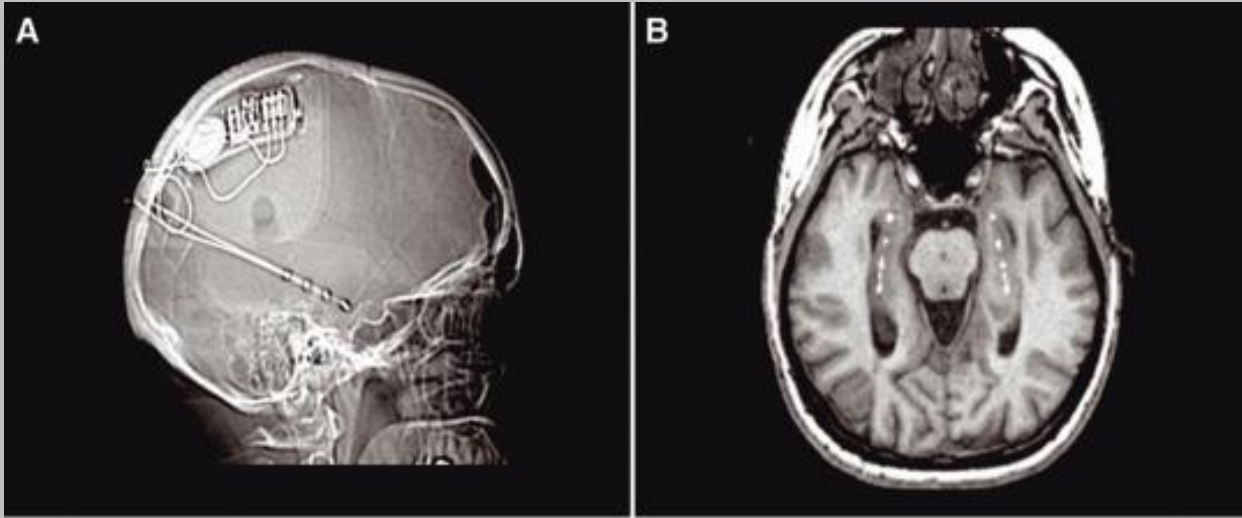


**NEW YORK INSTITUTE
OF TECHNOLOGY**

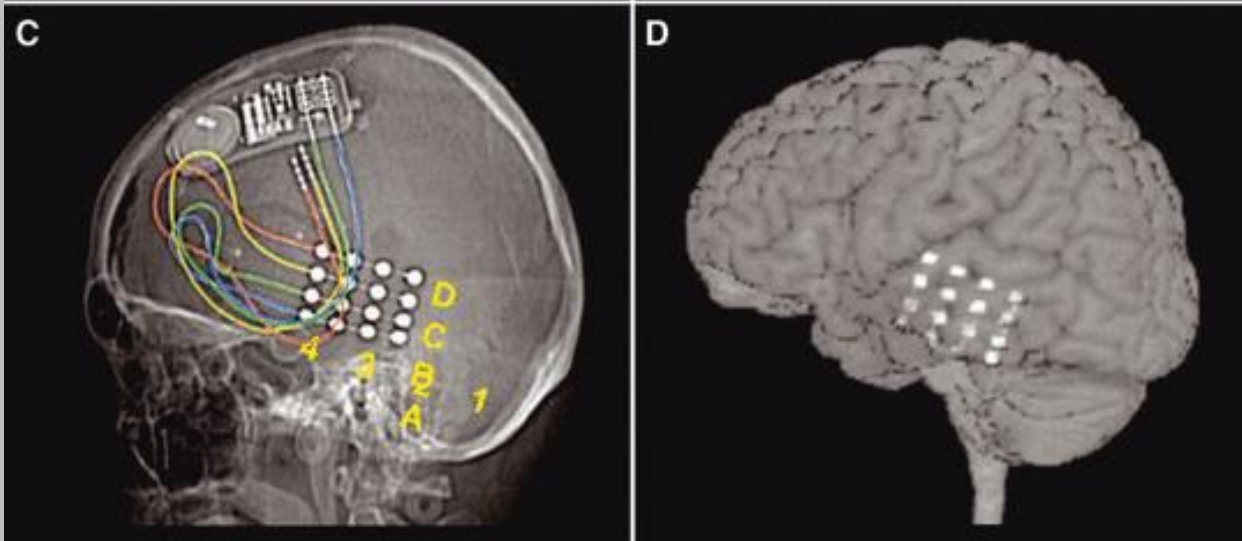
College of Osteopathic
Medicine

Stimulation of the hippocampus and temporal cortex for intractable epilepsy

Hippocampus - bilateral



Temporal cortex

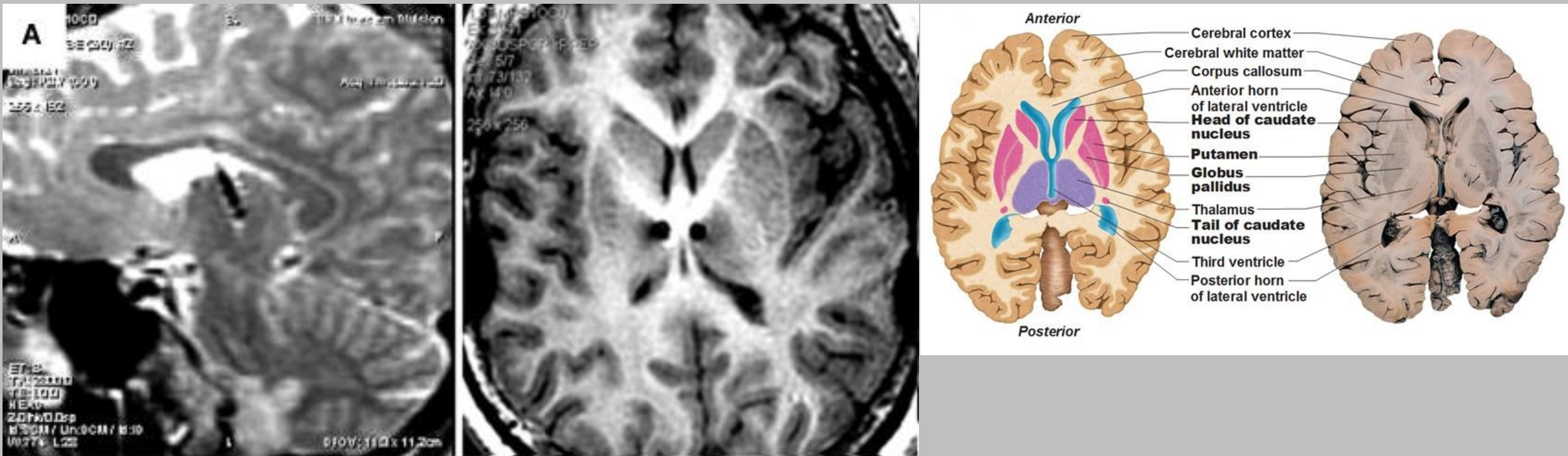


Similar approach to pace-maker for cardiac arrhythmia

Effective in some patients

Can be combined with other therapy modalities

Deep brain stimulation of the thalamus for intractable-epilepsy



Zumsteg D et al. 2006

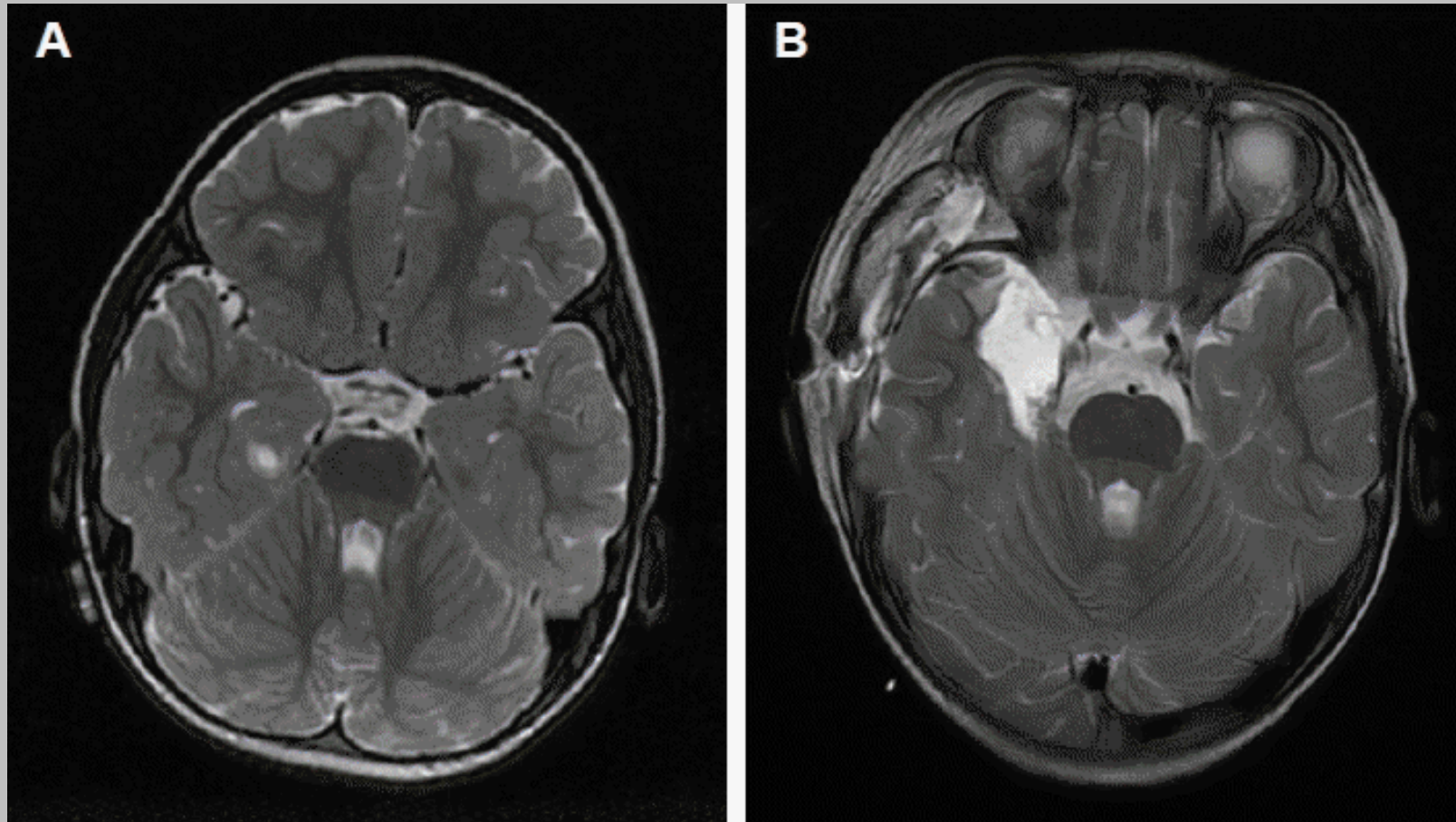
Similar approach to pace- maker for cardiac arrhythmia

Effective in some patients (50%+ reduction in seizures after 5-6yrs;
Andrade et al. 2006)

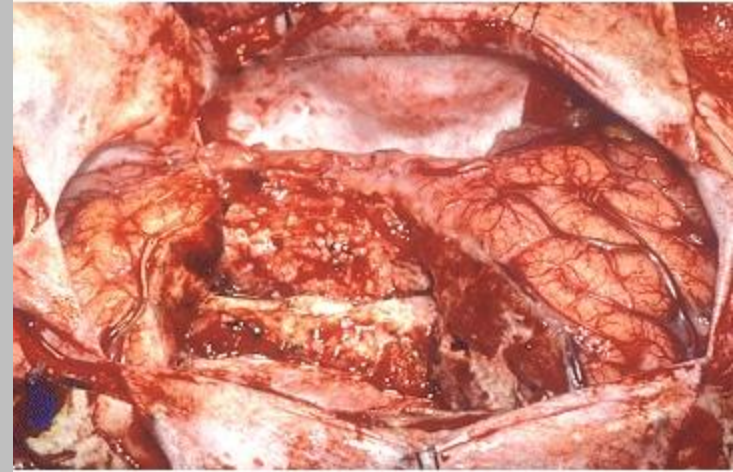
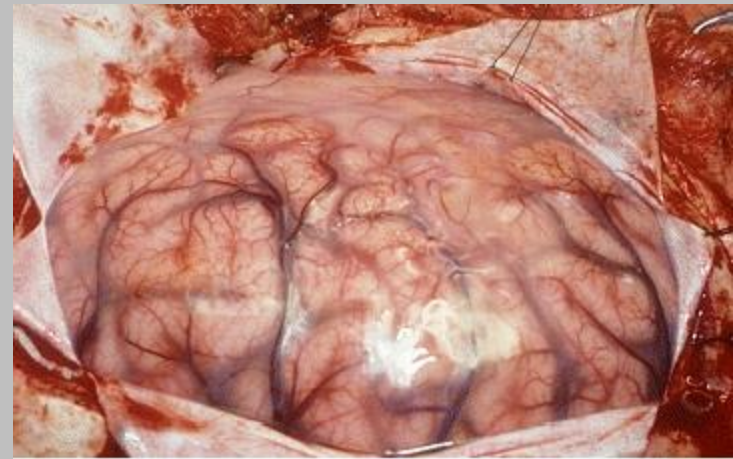
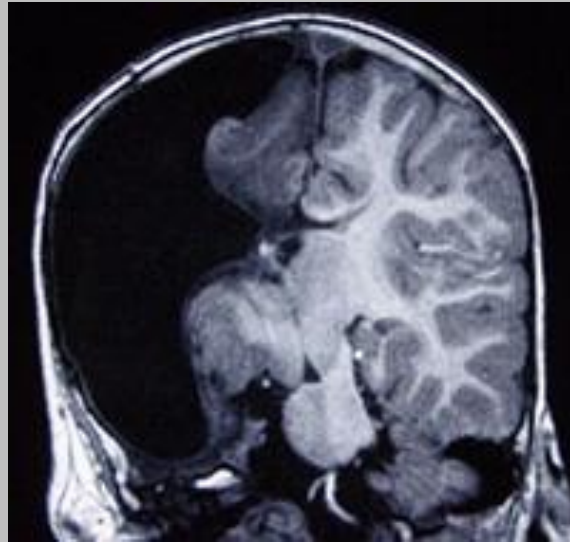
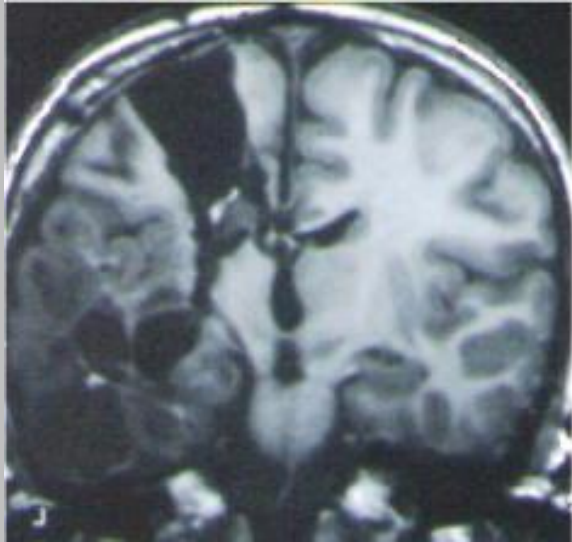
Can be combined with other therapy modalities

Surgical resection treatment of MTL epilepsy

- Goal – to destroy the abnormal circuits that initiate generalized seizures
- Challenge – to NOT destroy circuits important for other functions in activities of daily living.
- Last resort type of option



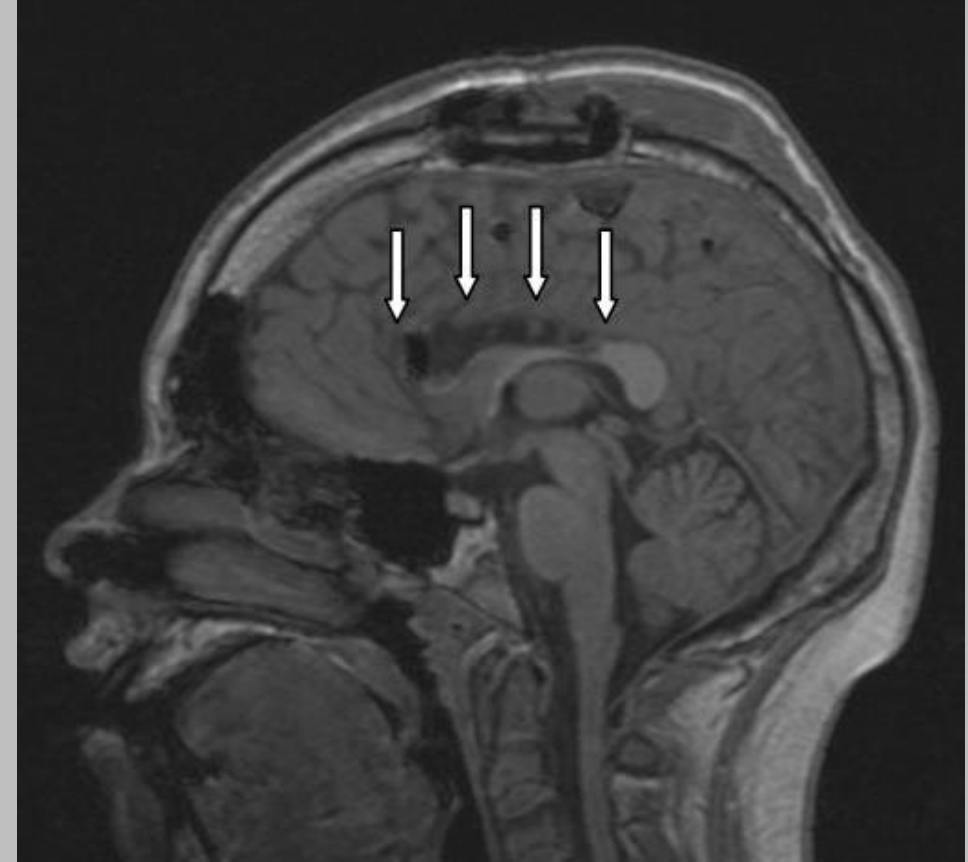
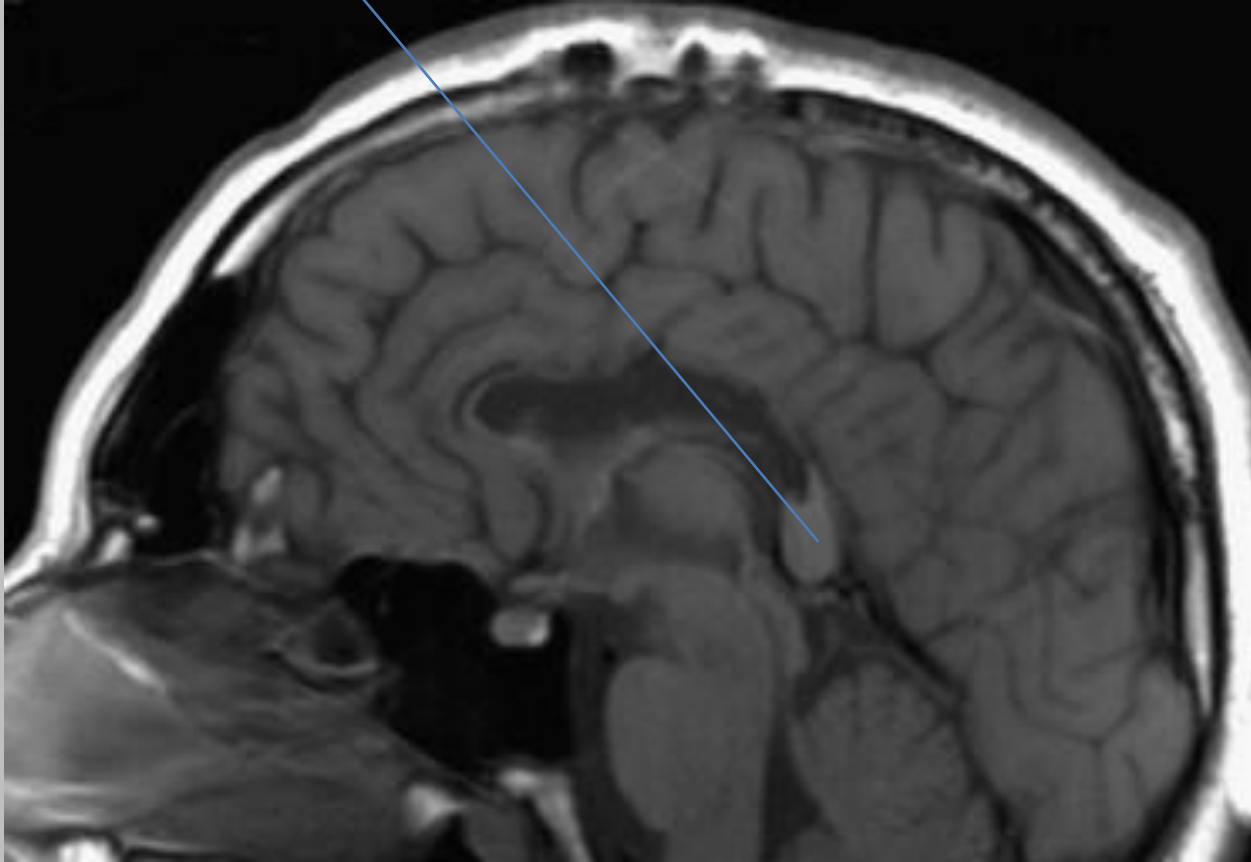
Hemispherectomy



- Patients with unilateral epilepsy that is poorly controlled with medication are candidates for this type of surgery.
- Given how invasive this is, would you expect this to be done in adults or peds more? And what difference accounts for this reason?

Callosotomy for the treatment of epilepsy

Splenium left intact



Goal: to limit epileptic activity to one hemisphere while preserving the callosal connections important for behavior. Prevent focal seizure from becoming a generalized seizure

Conclusions

- Seizures & epilepsy
 - Definitions: of seizure vs epilepsy. Seizure type ex) clonic vs tonic
 - Types of seizures: focal onset vs generalized.. Jacksonian March, febrile seizures etc..
 - Causes of epilepsy and example syndromes: Dravet's syndrome, Absence Epilepsy..
 - Treatments for epilepsy: Surgery, Deep Brain Stimulation or Vagus Nerve Stimulator among others

Question

- A 9 year-old pt is brought to his pediatrician for "spacing out" in both class and at home. During these episodes, the pt is seen abruptly stopping what he was doing. He then turns his head at an angle and look up at the ceiling for about 10 seconds. After this, the pt resumes what he was doing before the episode without any confusion or amnesia after the event. 1) What should the pediatrician do to help establish a diagnosis in this poor patient? 2) What is the drug used to treat this patient? 3) What is the theorized mechanism of action of the drug? 4) What major brain pathway or tract is affected by this condition?
- A) Hyperventilation of the pt, Levetiracetam, SV2 synaptic vesicle binding in presynaptic neuron, thalamocortical circuits
- B) Hypoventilation of the pt, ethosuximide, partially blockage of T-type calcium channels in thalamic neurons, corticospinal tract
- C) Hyperventilation of the pt, carbamazepine, blockage of presynaptic voltage gated Na^+ channels, Cerebral Peduncle
- D)Hypoventilation of the pt, carbamazepine, blockage of presynaptic voltage gated Na^+ channels, thalamocortical circuits
- E) Hyperventilation of the pt, ethosuximide, partially blockage of T-type calcium channels in thalamic neurons, thalamocortical circuits

Answer

E) Hyperventilation of the pt, ethosuximide, partially blockage of T-type calcium channels in thalamic neurons, thalamocortical circuits

Question

- A 75 year-old man with no past medical history presents to the ED with a complaint of abnormal twitching of his right upper arm for the past 2 minutes. The man said he was completely aware he was having a seizure and did not lose consciousness. The patient has no other neurological complaint and has had no neurological issues in the past. After CT and MRI was completed, EEG was positive for seizure activity. Which of the following is true about this patient?
 - A) This patient is having a generalized seizure with seizure activity in the motor cortex bilaterally
 - B) This patient is having a focal impaired awareness seizure in the motor cortex bilaterally
 - C) This patient is having a focal aware seizure in the left motor cortex
 - D) This patient is having a focal aware seizure in the right motor cortex
 - E) This patient is in status epilepticus and needs lorazepam

Answer

C) This patient is having a focal aware seizure in the left motor cortex

Question

Pt is 9 year old male who presents in the outpatient clinic after being admitted and treated for a seizure. The seizure lasted about 3 minutes and the pt's mother said they had no altered consciousness but a change in visual perception during the seizure. On physical exam, there are several hypopigmented skin patches on the neck of the patient. On the face of the patient, there are malformations made of fibrous tissue and blood vessels. What is one of the two genes involved in the pathogenesis of this disorder?

- A) GNAQ
- B) Hamartin (TSC1)
- C) Kir 4.1
- D) SCN1A
- E) Cacna1H

Answer

- Hamartin (TSC1)

Question

A 7-year-old boy with previous normal neuropsychiatric development presents with to the family med physician after being discharged 2 weeks ago from the hospital for a tonic-clonic seizure. About 6 months ago, the mother states her child is having trouble following commands and listening. Now the patient's mother states her child is not speaking as well as he previously did as is. EEG shows abnormalities in what correlates to Broca's and Wernicke's areas. What is the Diagnosis ad the gene implicated in the pathogenesis of the diagnosis?

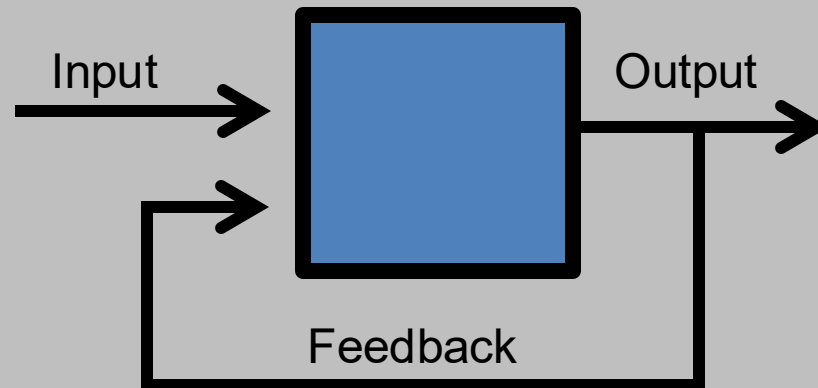
- A) Absence Seizures, **Cacna1H**
- B) EAST Syndrome, **KCNJ10**
- C) Sturge Weber Syndrome, **GNAQ**
- D) Autosomal dominant partial epilepsy with auditory features, **RELN**
- E) Lennox Gastaut syndrome, **GABRB3**

Answer

E) Lennox Gastaut syndrome, **GABRB3**

Feedback is welcomed

- <https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>



References

- Stuart M. Cain, Terrance P. Snutch, T-type calcium channels in burst-firing, network synchrony, and epilepsy, *Biochimica et Biophysica Acta (BBA) - Biomembranes*, Volume 1828, Issue 7, 2013, Pages 1572-1578, ISSN 0005-2736, <https://doi.org/10.1016/j.bbamem.2012.07.028>. (<https://www.sciencedirect.com/science/article/pii/S00527361200260X>)